

**Table 23.5** Properties of Selected Streptococci and Relatives

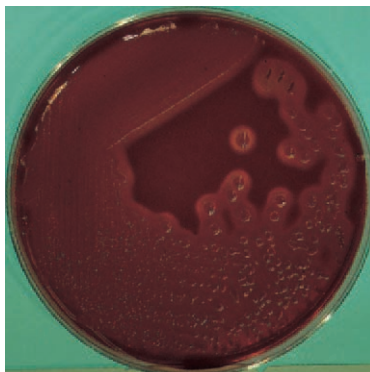
| Characteristics      | Pyogenic Streptococci | Oral Streptococci    |                   |                  | Enterococci        | Lactic Acid Streptococci |
|----------------------|-----------------------|----------------------|-------------------|------------------|--------------------|--------------------------|
|                      | <i>S. pyogenes</i>    | <i>S. pneumoniae</i> | <i>S. sanguis</i> | <i>S. mutans</i> | <i>E. faecalis</i> | <i>L. lactis</i>         |
| Growth at 10°C       | — <sup>a</sup>        | —                    | —                 | —                | +                  | +                        |
| Growth at 45°C       | —                     | —                    | d                 | d                | +                  | —                        |
| Growth at 6.5% NaCl  | —                     | —                    | —                 | —                | +                  | —                        |
| Growth at pH 9.6     | —                     | —                    | —                 | —                | +                  | —                        |
| Growth with 40% bile | —                     | —                    | d                 | d                | +                  | +                        |
| $\alpha$ -hemolysis  | —                     | +                    | +                 | —                | —                  | d                        |
| $\beta$ -hemolysis   | +                     | —                    | —                 | —                | +                  | —                        |
| Arginine hydrolysis  | +                     | +                    | +                 | —                | +                  | d                        |
| Hippurate hydrolysis | —                     | —                    | —                 | —                | +                  | d                        |
| Mol% G + C of DNA    | 35–39                 | 30–39                | 40–46             | 36–38            | 34–38              | 39                       |

Modified from *Bergey's Manual of Systematic Bacteriology*, Vol. 2, edited by P. H. A. Sneath, et al. Copyright © 1986 Williams and Wilkins, Baltimore, MD. Reprinted by permission.

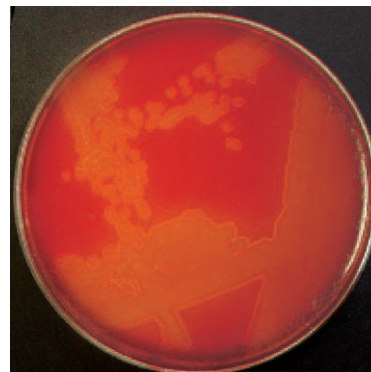
<sup>a</sup>Symbols: +, 90% or more of strains positive; —, 10% or less of strains positive; d, 11–89% of strains are positive.

**Figure 23.17** Streptococcal and Staphylococcal Hemolytic Patterns.

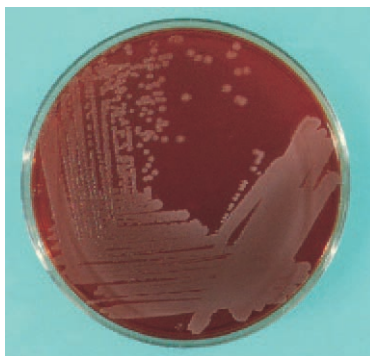
(a) *Streptococcus pyogenes* on blood agar, illustrating  $\beta$ -hemolysis. (b) *Streptococcus pneumoniae* on blood agar, illustrating  $\alpha$ -hemolysis. (c) *Staphylococcus aureus* on blood agar, illustrating  $\beta$ -hemolysis. (d) *Staphylococcus epidermidis* on blood agar with no hemolysis.



(a)



(b)



(c)



(d)

streptococcal colony;  **$\beta$ -hemolysis** is characterized by a zone of clearing or complete lysis without a marked color change. In addition, other hemolytic patterns are sometimes seen. Serological studies (see chapters 33 and 36) are also very important in identification because streptococci often have distinctive cell wall anti-

gens. Polysaccharide and teichoic acid antigens found in the wall or between the wall and the plasma membrane are used to identify these cocci, particularly pathogenic  $\beta$ -hemolytic streptococci, by the **Lancefield grouping system** (see Box 33.3). Biochemical and physiological tests are essential in identification (e.g., growth tem-

perature preferences, carbohydrate fermentation patterns, acetoin production, reduction of litmus milk, sodium chloride and bile salt tolerance, and the ability to hydrolyze arginine, esculin, hippurate, and starch). Sensitivity to bacitracin, sulfa drugs, and optochin (ethylhydrocuprein) also are used to identify particular species.

Members of the three genera have considerable practical importance. Pyogenic streptococci usually are pathogens and associated with pus formation (pyogenic means pus producing). Most species produce  $\beta$ -hemolysis on blood agar and form chains of cells. The major human pathogen in this group is *S. pyogenes* (streptococcal sore throat, acute glomerulonephritis, rheumatic fever). The normal habitat of oral streptococci is the oral cavity and upper respiratory tract of humans and other animals. In other respects oral streptococci are not necessarily similar. *S. pneumoniae* is  $\alpha$ -hemolytic and grows as pairs of cocci (figures 23.16c and 23.17b). It is associated with lobar pneumonia and otitis media (inflammation of the middle ear). *S. mutans* is associated with the formation of dental caries (see section 39.6). The enterococci such as *E. faecalis* are normal residents of the intestinal tracts of humans and most other animals. *E. faecalis* is an opportunistic pathogen that can cause urinary tract infections and endocarditis. Unlike the streptococci the enterococci will grow in 6.5% sodium chloride. The lactococci ferment sugars to lactic acid and can grow at 10°C but not at 45°C. *L. lactis* is widely used in the production of buttermilk and cheese (see sec-

tion 41.6) because it can curdle milk and add flavor through the synthesis of diacetyl and other products. [Streptococcal diseases \(pp. 903–6\)](#)

1. List the major properties of the genus *Bacillus*. What practical impacts does it have on society? Define parasporal body and Strickland reaction.
2. Briefly describe the genus *Thermoactinomyces*, with particular emphasis on its unique features. What disease does it cause?
3. What is distinctive about the morphology of *Caryophanon*?
4. Describe the genus *Staphylococcus*. How does the pathogen *S. aureus* differ from the common skin resident *S. epidermidis*, and where is it normally found?
5. List the major properties of the genus *Lactobacillus*. Why is it important in the food and dairy industries?
6. Describe the major distinguishing characteristics of the following taxa: *Streptococcus*, *Enterococcus*, *Lactococcus*, and *Leuconostoc*.
7. Of what practical importance is *Leuconostoc*? What are lactic acid bacteria?
8. What are  $\alpha$ -hemolysis,  $\beta$ -hemolysis, and the Lancefield grouping system?
9. Give a representative species and its importance for *Streptococcus*, *Enterococcus*, and *Lactococcus*. Distinguish between pyogenic and oral streptococci.

## Summary

1. Bacteria are placed in sections of volume 2 of the first edition of *Bergey's Manual* according to characteristics such as general shape, the possession of endospores, and their response to acid-fast staining.
2. Based on 16S rRNA analysis, gram-positive bacteria are divided into low G + C and high G + C groups; this system is used in the second edition of *Bergey's Manual*.
3. The second edition of *Bergey's Manual* places the low G + C gram positives in the phylum *Firmicutes*, which contains three classes: *Clostridia*, *Mollicutes*, and *Bacilli* (tables 23.1–23.3 and figure 23.1).
4. Mycoplasmas are gram-negative bacteria that lack cells walls and cannot synthesize peptidoglycan precursors. Many species require sterols for growth. They are the smallest bacteria capable of self-reproduction and usually grow on agar to give colonies a “fried-egg” appearance (figure 23.3).
5. Peptidoglycan structure often differs between groups in taxonomically useful ways. Most variations are in amino acid 3 of the peptide subunit or in the interpeptide bridge (figure 23.4).
6. Endospores resist desiccation and heat; they are used by bacteria to survive harsh conditions, especially in the soil.
7. Members of the genus *Clostridium* are anaerobic gram-positive rods that form endospores and don't carry out dissimilatory sulfate reduction (figure 23.6). They are responsible for botulism, tetanus, food spoilage, and putrefaction.
8. *Desulfotomaculum* is an anaerobic, endospore-forming genus that reduces sulfate to sulfide during anaerobic respiration.
9. The heliobacteria are gram-positive, anaerobic, photosynthetic bacteria with bacteriochlorophyll *g*. Some form endospores.
10. The family *Veillonellaceae* contains anaerobic, gram-negative cocci, some of which are parasites of vertebrates.
11. The class *Bacilli* is divided into two orders: *Bacillales* and *Lactobacillales*.
12. The genus *Bacillus* contains aerobic and facultative, catalase-positive, endospore-forming, chemoheterotrophic, gram-positive rods that are usually motile and peritrichous (figure 23.8). *Bacillus* synthesizes antibiotics and insecticides, and causes food poisoning and anthrax.
13. *Thermoactinomyces* is a gram-positive thermophile that forms a mycelium and true endospores (figure 23.10). It causes allergic reactions and leads to farmer's lung.
14. Members of the genus *Staphylococcus* are facultatively anaerobic, nonmotile, gram-positive cocci that form irregular clusters (figure 23.12). They grow on the skin and mucous membranes of warm-blood animals, and some are important human pathogens.
15. Several important genera such as *Lactobacillus*, *Listeria*, and *Caryophanon* contain regular, nonsporing, gram-positive rods. *Lactobacillus* carries out lactic acid fermentation and is extensively used in the food and dairy industries.
16. *Leuconostoc* carries out heterolactic fermentation using the phosphoketolase pathway (figure 23.15) and is involved in the production of fermented vegetable products, buttermilk, butter, and cheese.
17. The genera *Streptococcus*, *Enterococcus*, and *Lactococcus* contain gram-positive cocci arranged in pairs and chains that are usually facultative and carry out homolactic fermentation (tables 23.4 and 23.5). Some important species are the pyogenic coccus *S. pyogenes*, the oral streptococci *S. pneumoniae* and *S. mutans*, the enterococcus *E. faecalis* and lactococcus *L. lactis* (figure 23.16).

### Key Terms

$\alpha$ -hemolysis 531  
 $\beta$ -hemolysis 532  
coagulase 529

heterolactic fermentation 530  
lactic acid bacteria 529  
Lancefield grouping system 532

mycoplasmas 520  
parasporal body 525

### Questions for Thought and Review

1. On the basis of the treatment of gram-positive bacteria given in volume 2 of the first edition of *Bergey's Manual*, is it best to consider the volume as an identification guide, a description of phylogenetic relationships, or some of both? Explain your reasoning and conclusion.
2. Draw a diagram illustrating how gram-positive cocci might divide to produce the various clustering patterns observed (chains, tetrads, cubical packets, grapelike clusters).
3. Describe the characteristics most important in distinguishing between members of the following groups of genera: *Staphylococcus* and *Streptococcus*, *Bacillus* and *Clostridium*.
4. Which genera are noted for each of the following: coagulase, pathogenicity for insects, production of fermented foods and cheeses, use of the phosphoketolase pathway, causing allergic respiratory ailments, destruction of red blood cells, colonization of the tongue and teeth, and the generation of odors during putrefaction?
5. How might one go about determining whether the genome of *M. genitalium* is the smallest one compatible with a free-living existence?
6. Account for the ease with which anaerobic clostridia can be isolated from soil and other generally aerobic niches.

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### Critical Thinking Questions

1. Many low G + C bacteria are parasitic. The dependence on a host might be a consequence of the low G + C content. Elaborate on this concept.
2. The idea that *Bacillus* spores could be recovered from the intestines of bees preserved in amber is a bit *Jurassic Park*-like. Nevertheless, what types of controls would you wish to see performed in order to conclude that the spores were genuine and not contaminants from the current environment? (Assume the identification of the spore species was by PCR of rRNA sequences.)

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## CHAPTER 24

# Bacteria: The High G + C Gram Positives



*Frankia* forms nonmotile spores and is symbiotic with a number of higher plants such as alder trees.

### Outline

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- 24.4 Suborder *Micrococcineae* 542
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### Concepts

- 1. Volume 4 of the first edition of *Bergey's Manual* contains aerobic, gram-positive bacteria—the actinomycetes—that form branching hyphae and asexual spores.
- 2. The morphology and arrangement of spores, cell wall chemistry, and the types of sugars present in cell extracts are particularly important in actinomycete taxonomy and are used to divide these bacteria into different groups.
- 3. The second edition of *Bergey's Manual* classifies high G + C gram-positive bacteria using 16S rRNA data. They are placed in the phylum *Actinobacteria*, which contains the actinomycetes in volume 4 of the first edition plus gram-positive bacteria from sections 12, 15, 16, and 17 of volume 2.
- 4. Actinomycetes have considerable practical impact because they play a major role in the mineralization of organic matter in the soil and are the primary source of most naturally synthesized antibiotics. The genera *Corynebacterium* and *Mycobacterium* contain important human pathogens.



*Actinomycetes are very important from a medical point of view. . . . They may be a nuisance, as when they decompose rubber products, grow in aviation fuel, produce odorous substances that pollute water supplies, or grow in sewage-treatment plants where they form thick clogging foams. . . . In contrast, actinomycetes are the producers of most of the antibiotics.*

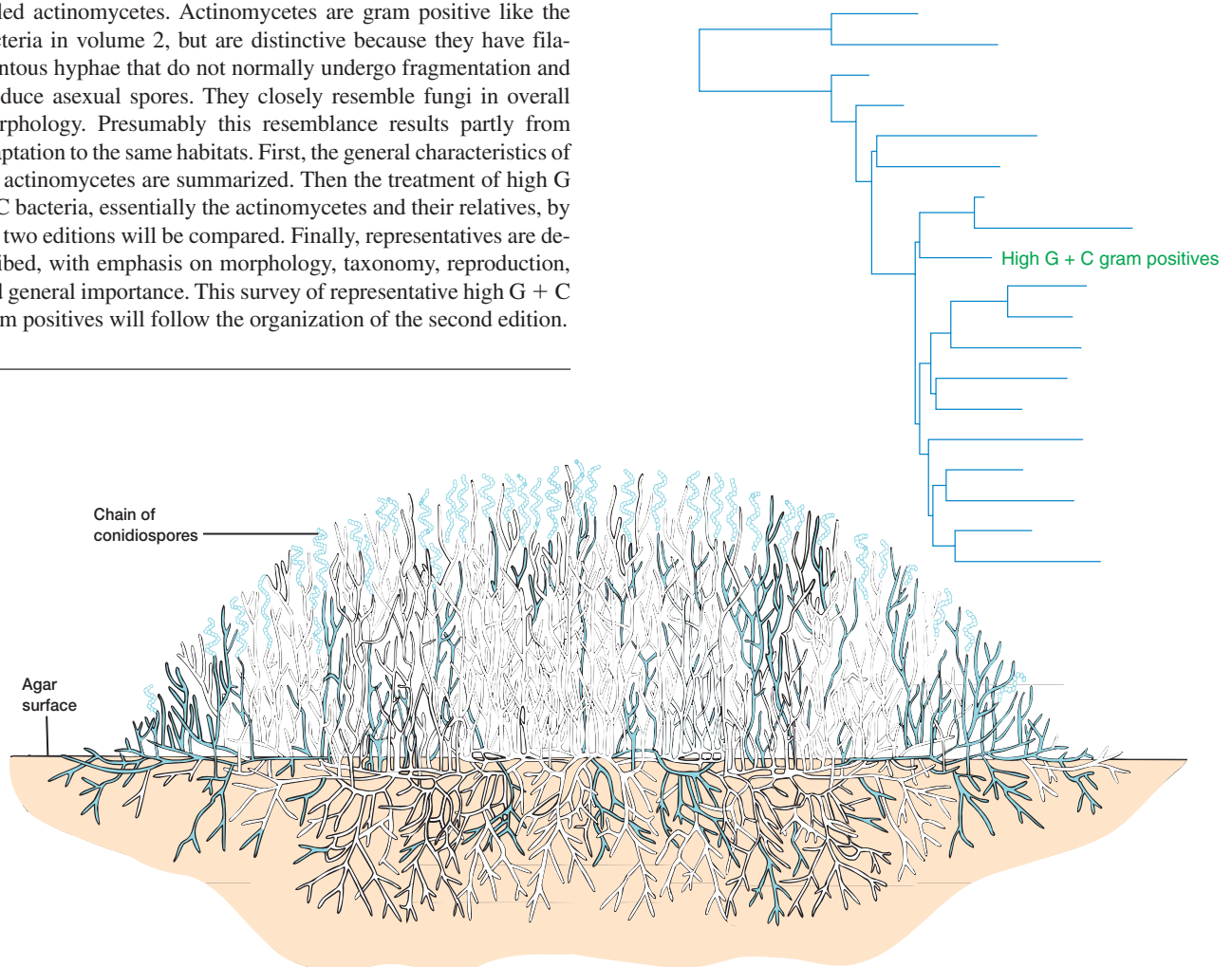
*H. A. Lechevalier and M. P. Lechevalier*

Chapter 24, the last of the survey chapters on bacteria, describes the high G + C gram-positive bacteria (see dendrogram on the right). These are located in volume 4 and part of volume 2 in the first edition of *Bergey's Manual of Systematic Bacteriology*; they will be in volume 4 of the second edition. The bacteria in volume 4 of the first edition are commonly called actinomycetes. Actinomycetes are gram positive like the bacteria in volume 2, but are distinctive because they have filamentous hyphae that do not normally undergo fragmentation and produce asexual spores. They closely resemble fungi in overall morphology. Presumably this resemblance results partly from adaptation to the same habitats. First, the general characteristics of the actinomycetes are summarized. Then the treatment of high G + C bacteria, essentially the actinomycetes and their relatives, by the two editions will be compared. Finally, representatives are described, with emphasis on morphology, taxonomy, reproduction, and general importance. This survey of representative high G + C gram positives will follow the organization of the second edition.

The **actinomycetes** [s., actinomycete] are aerobic, gram-positive bacteria that form branching filaments or hyphae and asexual spores. Although they are a diverse group, the actinomycetes do share many properties.

## 24.1 General Properties of the Actinomycetes

When growing on a solid substratum such as agar, the branching network of hyphae developed by actinomycetes grows both on the surface of the substratum and into it to form a substrate mycelium. Septa usually divide the hyphae into long cells (20  $\mu\text{m}$  and longer) containing several nucleoids. Sometimes a tissuelike mass results and may be called a **thallus**. Many actinomycetes also have an aerial mycelium that extends above the substratum and forms asexual, thin-walled spores called **conidia** [s., conidium] or **conidiospores** on the ends of filaments (**figure 24.1**). If the spores are located in a



**Figure 24.1 An Actinomycete Colony.** The cross section of an actinomycete colony with living (blue-green) and dead (white) hyphae. The substrate mycelium and aerial mycelium with chains of conidiospores are shown.

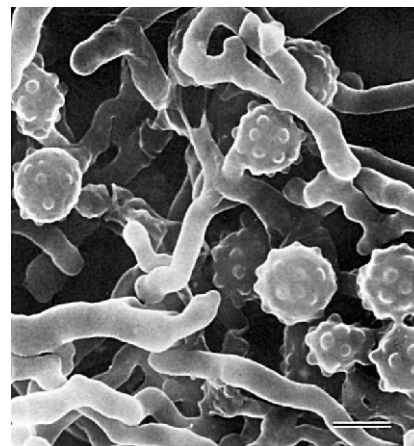
**Figure 24.2** Examples of Actinomycete Spores as Seen in the Scanning Electron Microscope. (a) Sporulating *Saccharopolyspora* hyphae (×3,000). (b) Sporangia of *Pilimelia columellifera* on mouse hair (× 520). (c) *Micromonospora echinospora*. Bar = 0.5 μm. (d) A chain of hairy streptomycete spores. Bar = 1.0 μm. (e) *Microbispora rosea*, paired spores on hyphae. (f) Aerial spores of *Kitasatospora setae*. Bar = 5 μm.



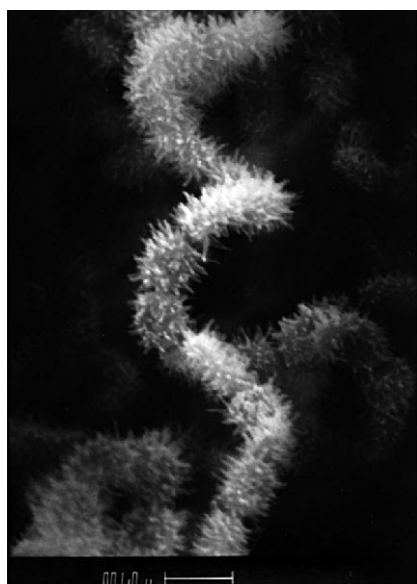
(a)



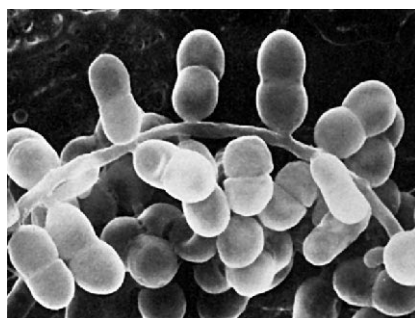
(b)



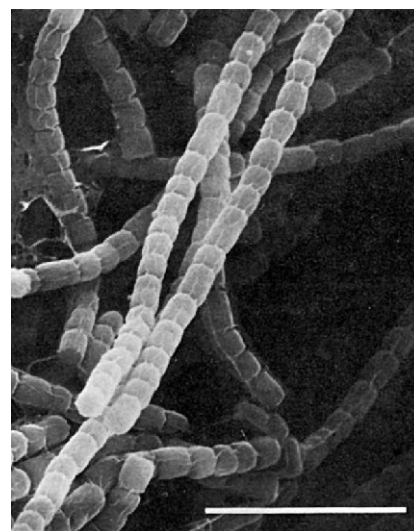
(c)



(d)



(e)



(f)

**Table 24.1** Actinomycete Cell Wall Types

| Cell Wall Type | Diaminopimelic Acid Isomer | Glycine in Interpeptide Bridge | Characteristic Sugars <sup>a</sup> | Representative Genera                                             |
|----------------|----------------------------|--------------------------------|------------------------------------|-------------------------------------------------------------------|
| I              | L, L                       | +                              | NA                                 | <i>Nocardioides</i> ,<br><i>Streptomyces</i> ,                    |
| II             | <i>meso</i>                | +                              | NA                                 | <i>Micromonospora</i> ,<br><i>Pilimelia</i> , <i>Actinoplanes</i> |
| III            | <i>meso</i>                | —                              | NA                                 | <i>Actinomadura</i> , <i>Frankia</i>                              |
| IV             | <i>meso</i>                | —                              | Arabinose, galactose               | <i>Saccharomonospora</i> ,<br><i>Nocardia</i>                     |

<sup>a</sup>NA, either not applicable or no diagnostic sugars.

sporangium, they are called **sporangiospores**. The spores can vary greatly in shape (**figure 24.2**). Actinomycete spores develop by septal formation at filament tips, usually in response to nutrient deprivation. Most are not particularly heat resistant but do withstand desiccation well and thus have considerable adaptive value.

Most actinomycetes are not motile. When motility is present, it is confined to flagellated spores.

Actinomycete cell wall composition varies greatly among different groups and is of considerable taxonomic importance. Four major cell wall types can be distinguished according to three features of peptidoglycan composition and structure: the amino acid in tetrapeptide side chain position 3, the presence of glycine in inter-

peptide bridges, and peptidoglycan sugar content (**table 24.1**). Cell extracts of actinomycetes with wall types II, III, and IV also contain characteristic sugars that are useful in identification (**table 24.2**). Some other taxonomically valuable properties are the morphology and color of mycelia and sporangia, the surface features and arrangement of conidiospores, the percent G + C in DNA, the phospholipid composition of cell membranes, and spore heat resistance. Newer techniques are being applied to actinomycete taxonomy. Comparison of 16S rRNA sequences has proven valuable as we shall see. Another useful technique is the production of large DNA fragments by restriction enzyme digestion and their separation and comparison using pulsed-field electrophoresis. [Gram-positive peptidoglycan structure and chemistry \(pp. 521–22\)](#)

Actinomycetes have considerable practical significance. They are primarily soil inhabitants and are very widely distributed. They can degrade an enormous number and variety of organic compounds and are extremely important in the mineralization of organic matter. Actinomycetes produce most of the medically useful natural antibiotics. Although most actinomycetes are free-living microorganisms, a few are pathogens of humans, other animals, and some plants.

## 24.2 High G + C Gram-Positive Bacteria in *Bergey's Manual*

The first edition of *Bergey's Manual* divides the actinomycetes into seven sections, primarily based on properties such as cell wall type, conidia arrangement, and the presence or absence of a sporangium (**table 24.3**).

It has been clear for some time that the sections in volume 4 of the first edition are not homogeneous and do not always fit with 16S rRNA sequence data. The second edition of *Bergey's Manual*

**Table 24.2** Actinomycete Whole Cell Sugar Patterns

| Sugar Pattern Types <sup>a</sup> | Characteristic Sugars | Representative Genera                                                   |
|----------------------------------|-----------------------|-------------------------------------------------------------------------|
| A                                | Arabinose, galactose  | <i>Nocardia</i> , <i>Rhodococcus</i> , <i>Saccharomonospora</i>         |
| B                                | Madurose <sup>b</sup> | <i>Actinomadura</i> , <i>Streptosporangium</i> , <i>Dermatophilus</i>   |
| C                                | None                  | <i>Thermomonospora</i> , <i>Actinosynnema</i> , <i>Geodermatophilus</i> |
| D                                | Arabinose, xylose     | <i>Micromonospora</i> , <i>Actinoplanes</i>                             |

<sup>a</sup>Characteristic sugar patterns are present only in wall types II–IV, those actinomycetes with *meso*-diaminopimelic acid.

<sup>b</sup>Madurose is 3-*O*-methyl-D-galactose

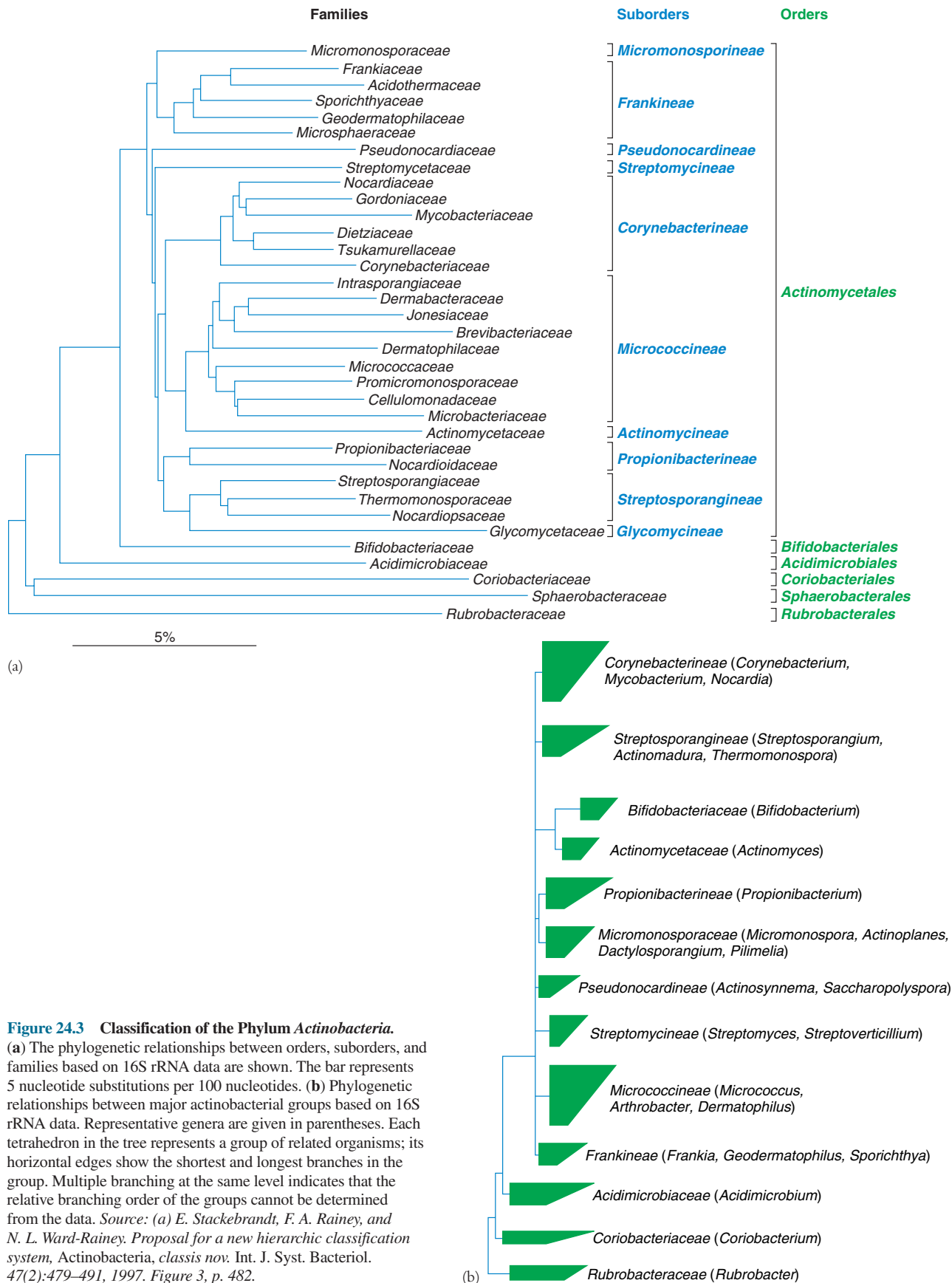
**Table 24.3** Some Characteristics of Actinomycete Groups in the First Edition of *Bergey's Manual*

| Group                                     | Wall Type              | Sugar Pattern          | Mol% G + C | Spore Arrangement                  | Presence of Sporangia | Selected Genera                                                                              |
|-------------------------------------------|------------------------|------------------------|------------|------------------------------------|-----------------------|----------------------------------------------------------------------------------------------|
| Nocardioform actinomycetes <sup>a</sup>   | I, IV, VI <sup>b</sup> | A                      | 59–79      | Varies                             | –                     | <i>Nocardia</i> , <i>Rhodococcus</i> , <i>Nocardioidea</i> , <i>Saccharomonospora</i>        |
| Actinomycetes with multilocular sporangia | III                    | B, C, D                | 57–75      | Clusters of spores                 | + (–) <sup>c</sup>    | <i>Geodermatophilus</i> , <i>Dermatophilus</i> , <i>Frankia</i>                              |
| Actinoplanetes                            | II                     | D                      | 71–73      | Varies                             | Usually +             | <i>Actinoplanes</i> , <i>Pilimelia</i> , <i>Dactylosporangium</i> , <i>Micromonospora</i>    |
| <i>Streptomyces</i> and related genera    | I                      | Not of taxonomic value | 69–78      | Chains of 5 to more than 50 spores | –                     | <i>Streptomyces</i> , <i>Sporichthya</i>                                                     |
| Maduromycetes                             | III                    | B, C                   | 64–74      | Varies                             | + or –                | <i>Actinomadura</i> , <i>Microbispora</i> , <i>Planomonospora</i> , <i>Streptosporangium</i> |
| <i>Thermomonospora</i> and related genera | III                    | C (sometimes B)        | 64–73      | Varies                             | –                     | <i>Thermomonospora</i> , <i>Actinosynnema</i> , <i>Nocardiopsis</i>                          |

<sup>a</sup>Several genera have mycolic acid. Filaments readily fragment into rods and coccoid elements.

<sup>b</sup>Type VI cell wall has lysine instead of diaminopimelic acid.

<sup>c</sup>Members have clusters of spores that may not always be surrounded by a sporangial wall.



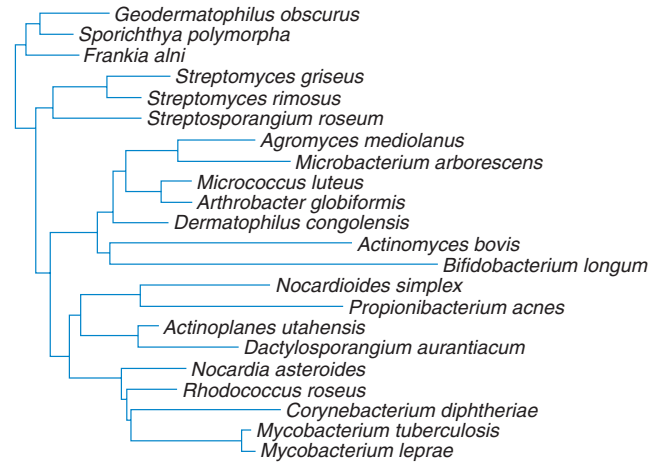
**Figure 24.3 Classification of the Phylum Actinobacteria.**

(a) The phylogenetic relationships between orders, suborders, and families based on 16S rRNA data are shown. The bar represents 5 nucleotide substitutions per 100 nucleotides. (b) Phylogenetic relationships between major actinobacterial groups based on 16S rRNA data. Representative genera are given in parentheses. Each tetrahedron in the tree represents a group of related organisms; its horizontal edges show the shortest and longest branches in the group. Multiple branching at the same level indicates that the relative branching order of the groups cannot be determined from the data. Source: (a) E. Stackebrandt, F. A. Rainey, and N. L. Ward-Rainey. Proposal for a new hierarchic classification system, Actinobacteria, classis nov. Int. J. Syst. Bacteriol. 47(2):479–491, 1997. Figure 3, p. 482.



uses 16S rRNA sequences to classify the high G + C gram positives, gram-positive bacteria with a DNA base composition above approximately 50 mol% G + C. Many of the genera from the first edition's sections 15 (irregular, nonsporing, gram-positive rods), 16 (the mycobacteria), and 17 (nocardioforms) are high G + C gram positives and are grouped with the actinomycetes in the second edition. Some members of the family *Micrococcaceae* (section 12) such as *Micrococcus* and *Stomatococcus* also are high G + C gram positives. All these bacteria are placed in the phylum *Actinobacteria* and classified as shown in **figure 24.3**. The phylum is large and very complex; it contains one class (*Actinobacteria*), five subclasses, six orders, 14 suborders, and 40 families. In this system the **actinobacteria** are composed of the actinomycetes and their high G + C relatives. The orders, suborders, and families are defined based on 16S rRNA sequences and distinctive signature nucleotides (*see p. 434*). **Figure 24.4** shows the phylogenetic relationships between selected representatives of the high G + C gram positives, and **table 24.4** summarizes the characteristics of some of the genera discussed in this chapter.

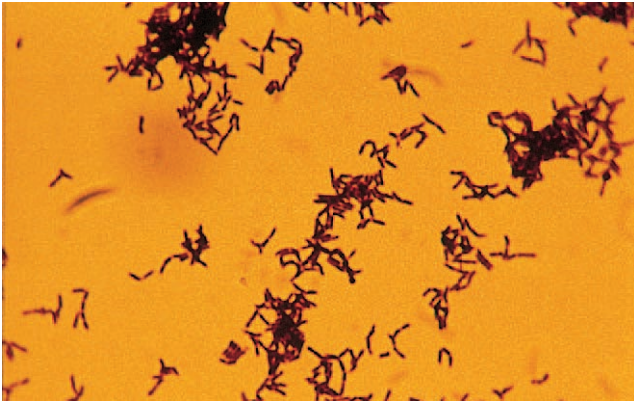
Most of the genera to be discussed in the following survey are in the subclass *Actinobacteridae* and order *Actinomycetales*



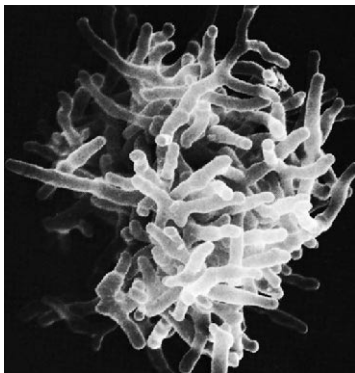
**Figure 24.4** Phylogenetic Relationships Among Selected High G + C Gram-Positive Bacteria. The relationships of a few species based on 16S rRNA sequence data are shown. *Source: The Ribosomal Database Project.*

**Table 24.4** Characteristics of Actinobacteria

| Genus                    | Dimensions (μm) and Morphology                                                                                                   | G + C Content (mol%) | Oxygen Requirement         | Other Distinctive Characteristics                                                                                                                        |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------|----------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Actinoplanes</i>      | Nonfragmenting, branching mycelium with little aerial growth; sporangia formed; motile spores with polar flagella                | 72–73                | Aerobic                    | Hyphae often in palisade arrangement; highly colored; type II cell walls; found in soil and decaying plant material                                      |
| <i>Arthrobacter</i>      | 0.8–1.2 × 1.0–8.0; young cells are irregular rods, older cells are small cocci; usually nonmotile                                | about 59–70          | Aerobic                    | Have rod-coccus growth cycle; metabolism respiratory; catalase positive; mainly in soil                                                                  |
| <i>Bifidobacterium</i>   | 0.5–1.3 × 1.5–8; rods of varied shape, usually curved; nonmotile                                                                 | 55–67                | Anaerobic                  | Cells can be clubbed or branched, pairs often in V arrangement; ferment carbohydrates to acetate and lactate, but no CO <sub>2</sub> ; catalase negative |
| <i>Corynebacterium</i>   | 0.3–0.8 × 1.5–8.0; straight or slightly curved rods with tapered or clubbed ends; nonmotile                                      | 51–63                | Facultatively anaerobic    | Cells often arranged in a V formation or in palisades of parallel cells; catalase positive and fermentative; metachromatic granules                      |
| <i>Frankia</i>           | 0.5–2.0 in diameter; vegetative hyphae with limited to extensive branching and no aerial mycelium; multilocular sporangia formed | 66–71                | Aerobic to microaerophilic | Sporangiospores nonmotile; usually fixes nitrogen; type III cell walls; most strains are symbiotic with angiosperm plants and induce nodules             |
| <i>Micrococcus</i>       | 0.5–2.0 diameter; cocci in pairs, tetrads, or irregular clusters; usually nonmotile                                              | 64–75                | Aerobic                    | Colonies usually yellow or red; catalase positive with respiratory metabolism; primarily on mammalian skin and in soil                                   |
| <i>Mycobacterium</i>     | 0.2–0.6 × 1.0–10; straight or slightly curved rods, sometimes branched; acid-fast; nonmotile and nonsporing                      | 62–70                | Aerobic                    | Catalase positive; can form filaments that are readily fragmented; walls have high lipid content; in soil and water; some parasitic                      |
| <i>Nocardia</i>          | 0.5–1.2 in diameter; rudimentary to extensive vegetative hyphae that can fragment into rod-shaped and coccoid forms              | 64–72                | Aerobic                    | Aerial hyphae formed; catalase positive; type IV cell wall; widely distributed in soil                                                                   |
| <i>Propionibacterium</i> | 0.5–0.8 × 1–5; pleomorphic nonmotile rods, may be bifid or branched; nonsporing                                                  | 53–67                | Anaerobic to aerotolerant  | Fermentation produces propionate and acetate, and often gas; catalase positive                                                                           |
| <i>Streptomyces</i>      | 0.5–2.0 in diameter; vegetative mycelium extensively branched; aerial mycelium forms chains of three to many spores              | 69–78                | Aerobic                    | Form discrete lichenoid, leathery, or butyrous colonies that often are pigmented; use many different organic compounds as nutrients; soil organisms      |



(a)



(b)

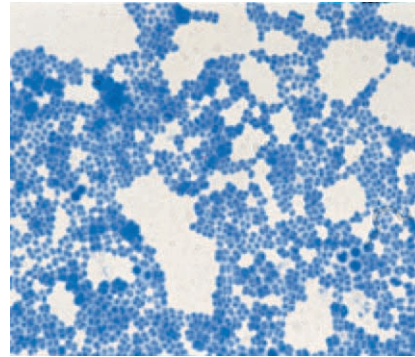
**Figure 24.5** Representatives of the Genus *Actinomyces*. (a) *A. naeslundii*; Gram stain ( $\times 1,000$ ). (b) *Actinomyces*; scanning electron micrograph ( $\times 18,000$ ). Note filamentous nature of the colony.

that is divided into 10 suborders. The following survey will focus on several of these suborders. The order *Bifidobacteriales* also will be briefly described.

### 24.3 Suborder *Actinomycineae*

Most genera have irregularly shaped, nonsporing, gram-positive rods with aerobic or facultative metabolism. The rods may be straight or slightly curved and usually have swellings, club shapes, or other deviations from normal rod morphology. *Actinomyces*, *Arcanobacterium*, and *Mobiluncus* are located in the suborder *Actinomycineae*.

Members of the genus *Actinomyces* are straight or slightly curved rods that vary considerably in shape and slender filaments with true branching (figure 24.5). The rods and filaments may have swollen, clubbed, or clavate ends. They are either facultative or strict anaerobes that require CO<sub>2</sub> for best growth. The cell walls contain lysine but not diaminopimelic acid or glycine. *Actinomyces* species are normal inhabitants of mucosal surfaces



**Figure 24.6** *Micrococcus*. *Micrococcus luteus*, methylene blue stain ( $\times 1,000$ ).

of humans and other warm-blooded animals; the oral cavity is their preferred habitat. *A. bovis* causes lumpy jaw in cattle. *Actinomyces* is responsible for actinomycoses, ocular infections, and periodontal disease in humans. The most important human pathogen is *A. israelii*.

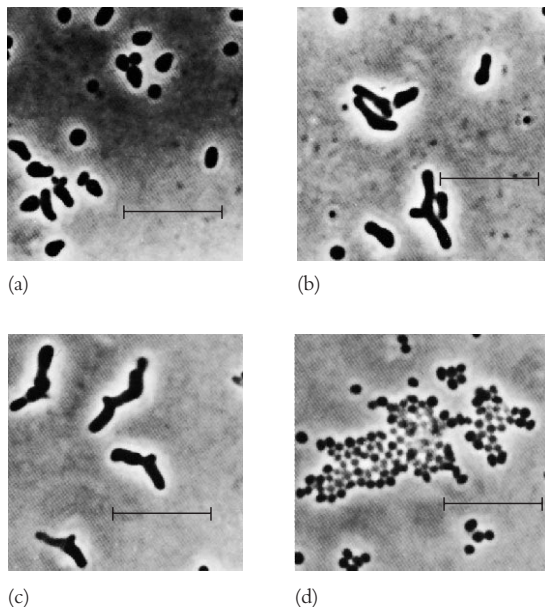
### 24.4 Suborder *Micrococcineae*

The suborder *Micrococcineae* has 10 families and a wide variety of genera. Two of the best-known genera are *Micrococcus* and *Arthrobacter*.

The genus *Micrococcus* contains aerobic, catalase-positive cocci that occur mainly in pairs, tetrads, or irregular clusters and are usually nonmotile (figure 24.6). Micrococci often are yellow, orange, or red in color. They are widespread in soil, water, and on mammalian skin; the last habitat may be their normal one. Despite their frequent presence on the skin, micrococci do not seem particularly pathogenic.

The genus *Arthrobacter* contains aerobic, catalase-positive rods with respiratory metabolism and lysine in its peptidoglycan. Its most distinctive feature is a rod-coccus growth cycle (figure 24.7). When *Arthrobacter* grows in exponential phase, the bacteria are irregular, branched rods that may reproduce by snapping division (see below). As they enter stationary phase, the cells change to a coccoid form. Upon transfer to fresh medium, the coccoid cells produce outgrowths and again form actively growing rods. Although arthrobacters often are isolated from fish, sewage, and plant surfaces, their most important habitat is the soil, where they constitute a significant component of the microbial flora. They are well adapted to this niche because they are very resistant to desiccation and nutrient deprivation. This genus is unusually flexible nutritionally and can even degrade some herbicides and pesticides; it is probably important in the mineralization of complex organic molecules.

The mechanism of **snapping division** has been studied in *Arthrobacter*. These bacteria have a two-layered cell wall, and only the inner layer grows inward to generate a transverse wall di-



**Figure 24.7 The Rod-Coccus Growth Cycle.** The rod-coccus cycle of *Arthrobacter globiformis* when grown at 25°C. (a) Rods are outgrowing from cocci 6 hours after inoculation. (b) Rods after 12 hours of incubation. (c) Bacteria after 24 hours. (d) Cells after reaching stationary phase (3 days' incubation). The cells used for inoculation resembled these stationary-phase cocci. Bars = 10  $\mu$ m.

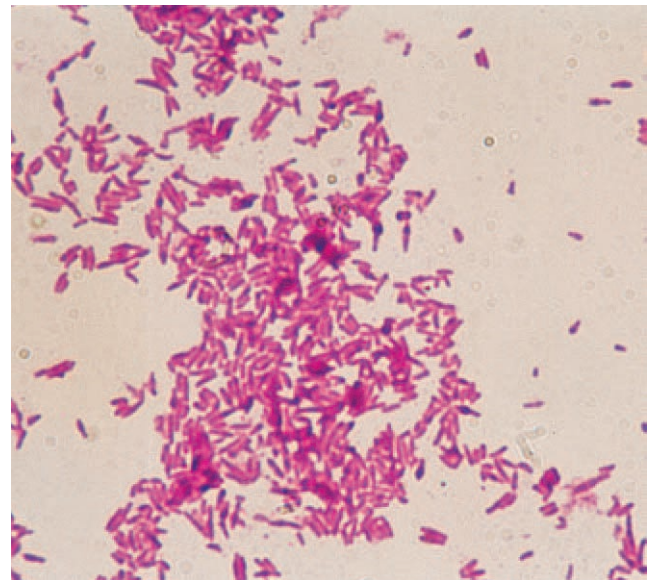
viding the new cells. The completed transverse wall or septum next thickens and puts tension on the outer wall layer, which still holds the two cells together. Eventually, increasing tension ruptures the outer layer at its weakest point, and a snapping movement tears the outer layer apart around most of its circumference. The new cells now rest at an angle to each other and are held together by the remaining portion of the outer layer that acts as a hinge.

A third genus in this suborder is *Dermatophilus*. *Dermatophilus* (cell wall type IIIB) also forms packets of motile spores with tufts of flagella, but it is a facultative anaerobe and a parasite of mammals responsible for the skin infection streptothrichosis.

## 24.5 Suborder *Corynebacterineae*

This suborder contains seven families with several well-known genera. Three of the most important genera are *Corynebacterium*, *Mycobacterium*, and *Nocardia*.

The family *Corynebacteriaceae* contains one genus, *Corynebacterium*, which contains aerobic and facultative, catalase-positive, straight to slightly curved rods, often with tapered ends. Club-shaped forms are also seen. The bacteria often remain partially attached after snapping division, resulting in angular arrangements of the cells, somewhat like Chinese letters, or a palisade arrangement in which rows of cells are lined up side by side (figure 24.8). *Corynebacteria* form metachromatic granules, and



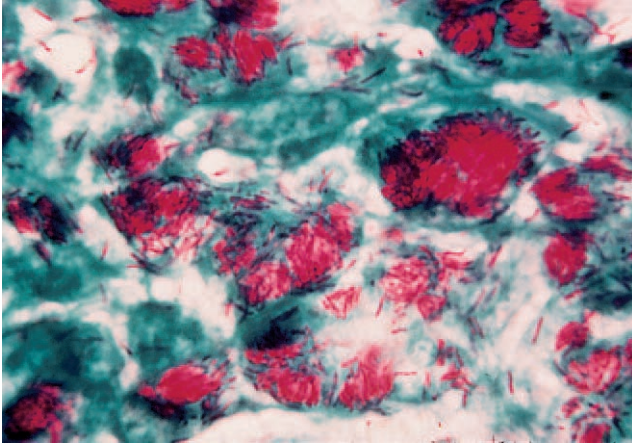
**Figure 24.8 *Corynebacterium diphtheriae*.** Note the irregular shapes of individual cells, the angular associations of pairs of cells, and palisade arrangements ( $\times 1,000$ ). These gram-positive rods do not form endospores.

their walls have *meso*-diaminopimelic acid. Although some species are harmless soil and water saprophytes, many corynebacteria are plant or animal pathogens. For example, *C. diphtheriae* is the causative agent of diphtheria in humans (see section 39.1).

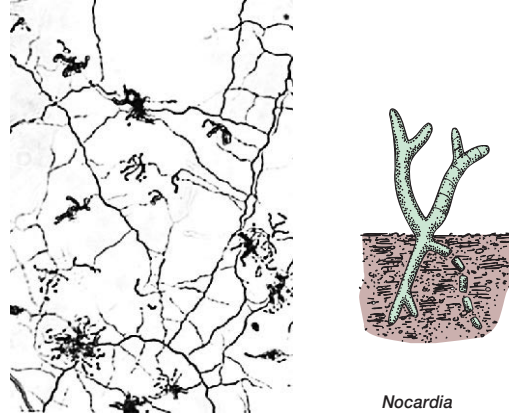
The family *Mycobacteriaceae* contains the genus *Mycobacterium*, which is composed of slightly curved or straight rods that sometimes branch or form filaments (figure 24.9). Mycobacterial filaments differ from those of actinomycetes in readily fragmenting into rods and coccoid bodies when distributed. They are aerobic and catalase positive. Mycobacteria grow very slowly and must be incubated for 2 to 40 days after inoculation of a solidified complex medium to form a visible colony. Their cell walls have a very high lipid content and contain waxes with 60 to 90 carbon **mycolic acids**. These are complex fatty acids with a hydroxyl group on the  $\beta$ -carbon and an aliphatic chain attached to the  $\alpha$ -carbon. The presence of mycolic acids and other lipids outside the peptidoglycan layer makes mycobacteria **acid-fast** (basic fuchsin dye cannot be removed from the cell by acid-alcohol treatment). Extraction of wall lipid with alkaline ethanol destroys acid-fastness. *Actinomycete morphology* (pp. 537–39); *Acid-fast staining* (p. 28)

Although some mycobacteria are free-living saprophytes, they are best known as animal pathogens. *M. bovis* causes tuberculosis in cattle, other ruminants, and primates. Because this bacterium can produce tuberculosis in humans, dairy cattle are tested for the disease yearly; milk pasteurization kills the pathogen and affords further protection against disease transmission. Thus *M. tuberculosis* is the chief source of tuberculosis in humans. The





**Figure 24.9 The Mycobacteria.** *Mycobacterium leprae*. Acid-fast stain ( $\times 400$ ). Note the masses of red mycobacteria within blue-green host cells.



**Figure 24.10 Nocardia.** *Nocardia asteroides*, substrate mycelium and aerial mycelia with conidia illustration and light micrograph ( $\times 1,250$ ).

other major mycobacterial human disease is leprosy, caused by *M. leprae*. [Tuberculosis](#) (pp. 906–8); [Leprosy](#) (pp. 916–17); [The genome sequences of \*M. tuberculosis\* and \*M. leprae\*](#) (pp. 351–52)

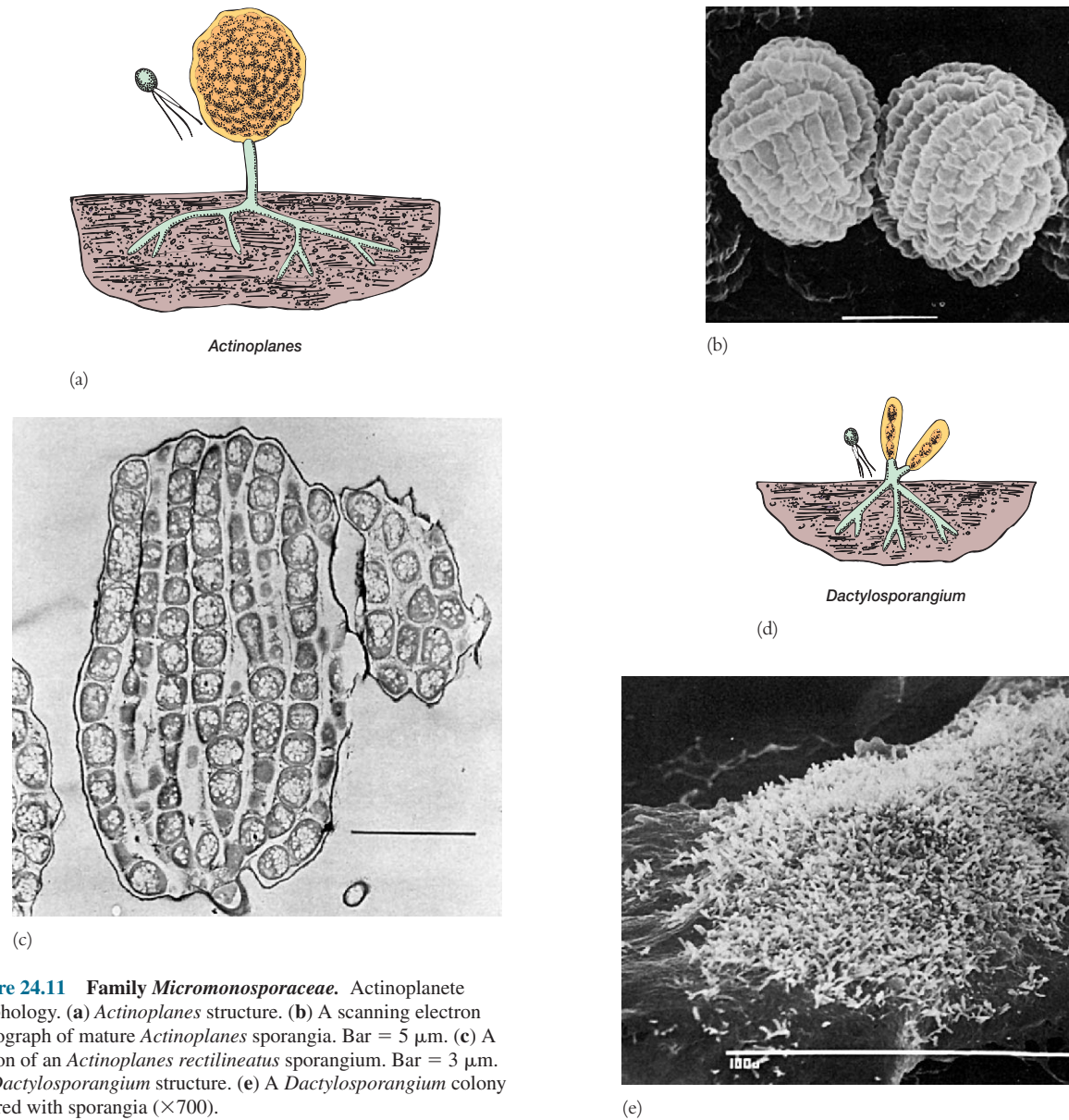
The family *Nocardiaceae* is composed of two genera, *Nocardia* and *Rhodococcus*. Because these and related genera resemble members of the genus *Nocardia* (named after Edmond Nocard [1850–1903], French bacteriologist and veterinary pathologist), they are collectively called **nocardioforms**. These bacteria develop a substrate mycelium that readily breaks into rods and coccoid elements (**figure 24.10**). Several genera also form an aerial mycelium that rises above the substratum and may produce conidia. All genera have a high G + C content like other actinomycetes, and almost all are strict aerobes. Most species have peptidoglycan with *meso*-diaminopimelic acid and no peptide interbridge. The wall usually contains a carbohydrate composed of arabinose and galactose; mycolic acids are present in *Nocardia* and *Rhodococcus*.

*Nocardia* is distributed worldwide in soil and also is found in aquatic habitats. Nocardiae are involved in the degradation of hydrocarbons and waxes and can contribute to the biodeterioration of rubber joints in water and sewage pipes. Although most are free-living saprophytes, some species, particularly *N. asteroides*, are opportunistic pathogens that cause nocardiosis in humans and other animals. People with low-resistance due to other health problems are most at risk. The lungs are most often infected, but the central nervous system and other organs may be invaded. *Rhodococcus* also is widely distributed in soils and aquatic habitats. It is of considerable interest because members of the genus can degrade an enormous variety of molecules such as petroleum hydrocarbons, detergents, benzene, polychlorinated biphenyls (PCBs), and various pesticides. It may be possible to use rhodococci to remove sulfur from fuels, thus reducing air pollution by sulfur oxide emissions.

1. Define actinomycete, thallus, substrate mycelium, aerial mycelium, conidium, conidiospore, and sporangiospore.
2. Describe how cell wall structure and sugar content are used to classify the actinomycetes. Include a brief description of the four major wall types.
3. Why are the actinomycetes of such practical interest?
4. How do the first and second editions of *Bergey's Manual* differ in their treatment of the high G + C gram-positive bacteria?
5. Give the major distinguishing characteristics of the actinomycetes in the first edition of *Bergey's Manual*. Describe the phylum *Actinobacteria* and its relationship to the actinomycetes.
6. Describe the major characteristics of the following genera: *Actinomyces*, *Micrococcus*, *Arthrobacter*, and *Corynebacterium*. Include comments on their normal habitat and importance.
7. What is snapping division? The rod-coccus growth cycle?
8. Give the distinctive properties of the genus *Mycobacterium*, and indicate how mycobacteria differ from actinomycetes.
9. Define mycolic acid and acid-fast.
10. List two human mycobacterial diseases and their causative agents. Which pathogen causes tuberculosis in cattle?
11. What is a nocardioform, and how can the group be distinguished from other actinomycetes?
12. Where is *Nocardia* found, and what problems may it cause? Consider both environmental and public health concerns.

## 24.6 Suborder *Micromonosporineae*

The suborder *Micromonosporineae* contains only one family, *Micromonosporaceae*. Many of the genera in this family are placed among the actinoplanetes in the first edition of *Bergey's Manual*. Actinoplanetes [Greek *actinos*, a ray or beam, and *planes*, a wan-

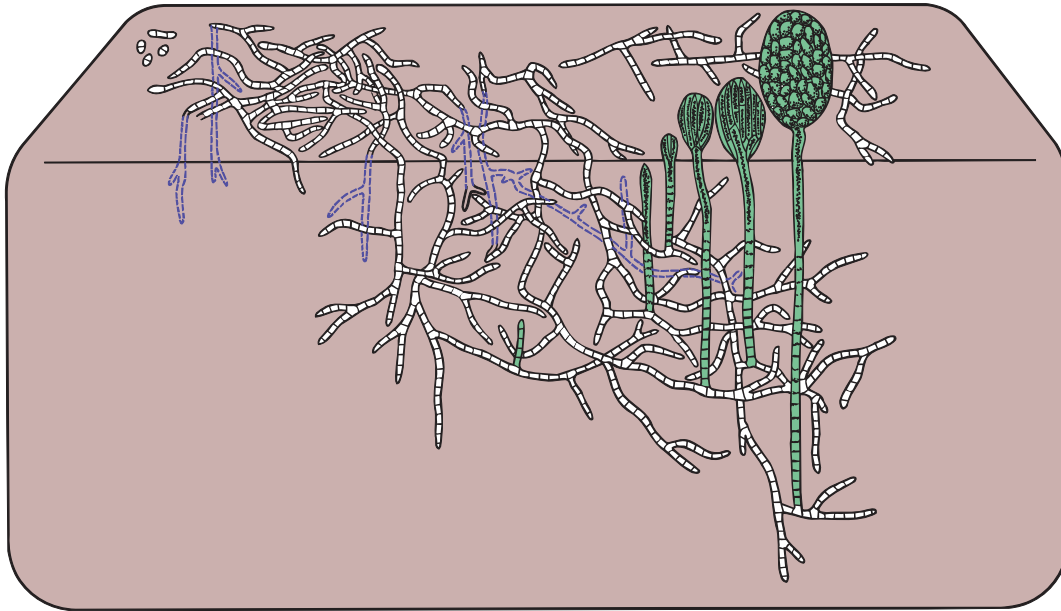


**Figure 24.11** Family *Micromonosporaceae*. Actinoplanete morphology. (a) *Actinoplanes* structure. (b) A scanning electron micrograph of mature *Actinoplanes* sporangia. Bar = 5  $\mu\text{m}$ . (c) A section of an *Actinoplanes rectilineatus* sporangium. Bar = 3  $\mu\text{m}$ . (d) *Dactylosporangium* structure. (e) A *Dactylosporangium* colony covered with sporangia ( $\times 700$ ).

derer] have an extensive substrate mycelium and are wall type IID. Often the hyphae are highly colored and diffusible pigments may be produced. Normally an aerial mycelium is absent or rudimentary. Conidiospores are usually formed within a sporangium raised above the surface of the substratum at the end of a special hypha called a sporangiophore. The spores can be either motile or nonmotile. These bacteria vary in the arrangement and development of their spores. Some genera (*Actinoplanes*, *Pilimelia*) have spherical, cylindrical, or irregular sporangia with a few to several thousand spores per sporangium (figure 24.2b and figure 24.11).

The sporangium develops above the substratum at the tip of a sporangiophore; the spores are arranged in coiled or parallel chains (figure 24.12). *Dactylosporangium* forms club-shaped, finger-like, or pyriform sporangia with one to six spores (figure 24.11d,e). *Micromonospora* bears single spores, which often occur in branched clusters of sporophores (figure 24.2c).

Actinoplanetes [s., actinoplanete] grow in almost all soil habitats, ranging from forest litter to beach sand. They also flourish in freshwater, particularly in streams and rivers (probably because of abundant oxygen and plant debris). Some have been isolated from



**Figure 24.12** Sporangium Development in an Actinoplanete. The developing sporangium is shown in green.

the ocean. The soil-dwelling species may have an important role in the decomposition of plant and animal material. *Pilmelia* grows in association with keratin. *Micromonospora* actively degrades chitin and cellulose, and it can produce antibiotics such as gentamicin.

### 24.7 Suborder *Propionibacterineae*

This suborder contains two families and 10 genera. The genus *Propionibacterium* will be placed in the family *Propionibacteriaceae* in the second edition. The genus contains pleomorphic, nonmotile, nonsporing rods that are often club-shaped with one end tapered and the other end rounded. Cells also may be coccoid or even branched. They can be single, in short chains, or in clumps. The genus is facultatively anaerobic or aerotolerant; lactate and sugars are fermented to produce large quantities of propionic and acetic acids, and often carbon dioxide. *Propionibacterium* is usually catalase positive. The genus is found growing on the skin and in the digestive tract of animals, and in dairy products such as cheese. *Propionibacterium* contributes substantially to the production of Swiss cheese (see p. 982). *P. acnes* is involved with the development of body odor and acne vulgaris (see p. 701).

### 24.8 Suborder *Streptomycineae*

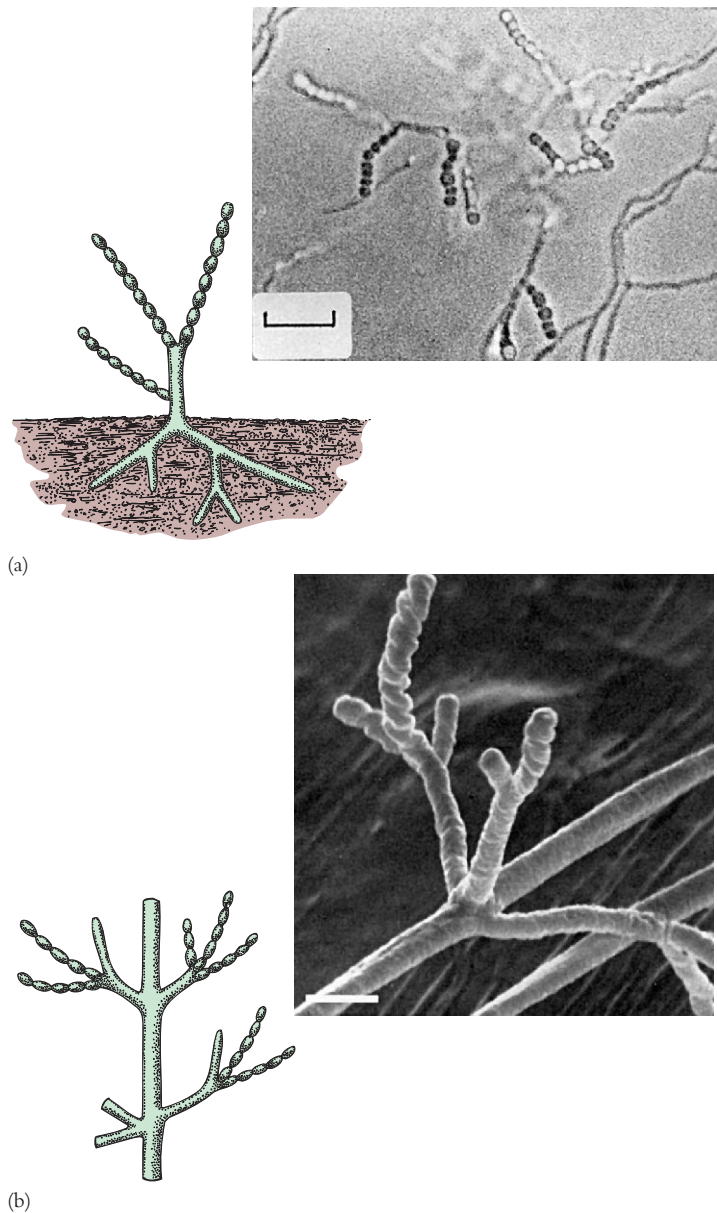
The suborder *Streptomycineae* has only one family, *Streptomyetaceae*, and three genera, the most important of which is *Streptomyces*. The first edition of *Bergey's Manual* places *Streptomyces* in section 29 with genera whose aerial hyphae divide in a

single plane to form chains of 3 to 50 or more nonmotile conidiospores with surface texture ranging from smooth to spiny and warty (figure 24.2d; figures 24.13 and 24.14). All have a type I cell wall and a G + C content of around 69 to 78%. The substrate mycelium, when present, does not undergo fragmentation. Members of the section are often called **streptomycetes** [Greek *streptos*, bent or twisted, and *myces*, fungus].

*Streptomyces* is an enormous genus; there are around 500 species. Members of the genus are strict aerobes, are wall type I, and form chains of nonmotile spores within a thin, fibrous sheath (figure 24.14). The three to many conidia in each chain are often pigmented and can be smooth, hairy, or spiny in texture. *Streptomyces* species are determined by means of a mixture of morphological and physiological characteristics, including the following: the color of the aerial and substrate mycelia, spore arrangement, surface features of individual spores, carbohydrate use, antibiotic production, melanin synthesis, nitrate reduction, and the hydrolysis of urea and hippuric acid.

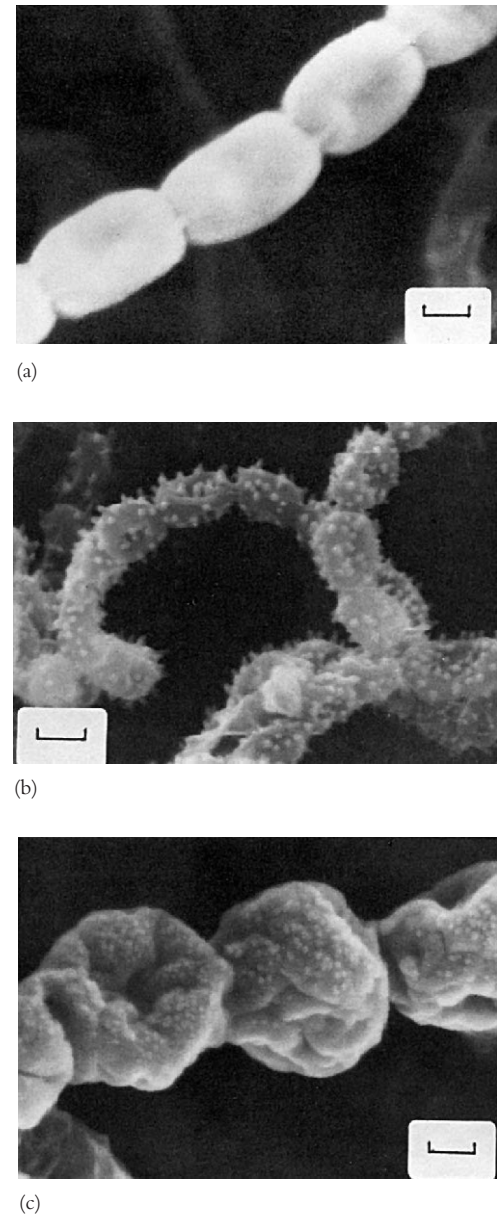
Streptomycetes are very important, both ecologically and medically. The natural habitat of most streptomycetes is the soil, where they may constitute from 1 to 20% of the culturable population. In fact, the odor of moist earth is largely the result of streptomycete production of volatile substances such as **geosmin**. Streptomycetes play a major role in mineralization. They are flexible nutritionally and can aerobically degrade resistant substances such as pectin, lignin, chitin, keratin, latex, and aromatic compounds. Streptomycetes are best known for their synthesis of a vast array of antibiotics, some of which are useful in medicine and biological research. Examples include amphotericin B, chloramphenicol, erythromycin, neomycin, nystatin, streptomycin,





**Figure 24.13** *Streptomyces* and Related Genera. Streptomycete conidia arrangement. (a) An illustration of typical *Streptomyces* morphology; a light micrograph of *S. carpinensis* spore chains. Bar = 5  $\mu\text{m}$ . (b) *Streptoverticillium* (*Streptomyces*) morphology; a scanning electron micrograph of *Sv. salmonis* with developing spore chains. Bar = 2  $\mu\text{m}$ .

tetracycline, and so forth (figure 24.15a). Although most streptomycetes are nonpathogenic saprophytes, a few are associated with plant and animal diseases. *Streptomyces scabies* causes scab disease in potatoes and beets (figure 24.15b). *S. somaliensis* is the only streptomycete known to be pathogenic for humans. It is associated with **actinomycetoma**, an infection of subcutaneous tissues that produces lesions and leads to swelling, abscesses, and

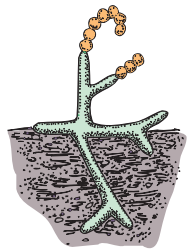
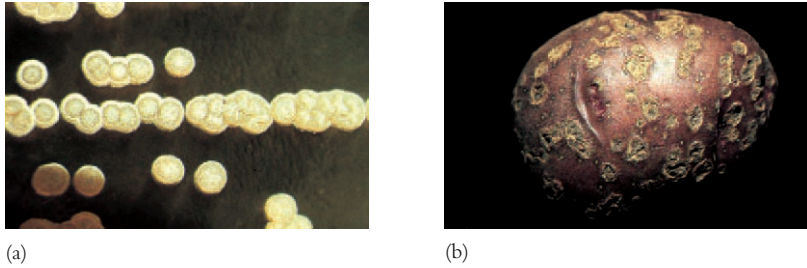


**Figure 24.14** *Streptomycete* Spores. (a) Smooth spores of *S. niveus*; scanning electron micrograph. Bar = 0.25  $\mu\text{m}$ . (b) Spiny spores of *S. viridochromogenes*. Bar = 0.5  $\mu\text{m}$ . (c) Warty spores of *S. pulcher*. Bar = 0.25  $\mu\text{m}$ .

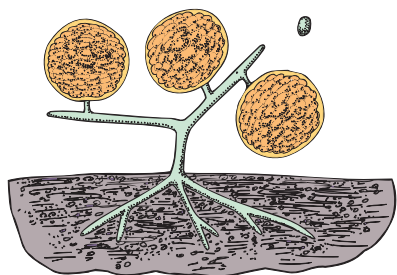
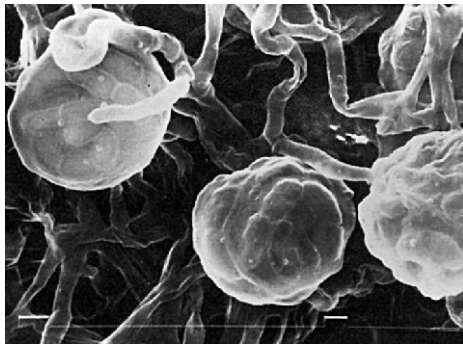
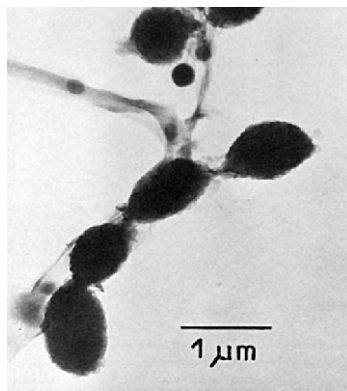
even bone destruction if untreated. *S. albus* and other species have been isolated from patients with various ailments and may be pathogenic. [Antibiotics and their properties \(chapter 35\)](#)

*Streptoverticillium*, another genus in the family, has an aerial mycelium with a whorl of three to six short branches that are produced at fairly regular intervals. These branches have secondary branches bearing chains of spores (figure 24.13b).

**Figure 24.15 Streptomycetes of Practical Importance.** (a) *Streptomyces griseus*. Colonies of the actinomycete that produces streptomycin. (b) *Streptomyces scabies*, an actinomycete growing on a potato.



*Actinomadura madurae*  
(a)



*Streptosporangium*  
(b)

**Figure 24.16 Maduromycetes.** (a) *Actinomadura madurae* morphology; illustration and electron micrograph of a spore chain ( $\times 16,500$ ). (b) *Streptosporangium* morphology; illustration and micrograph of *S. album* on oatmeal agar with sporangia and hyphae; SEM. Bar interval = 10  $\mu\text{m}$ .

## 24.9 Suborder Streptosporangineae

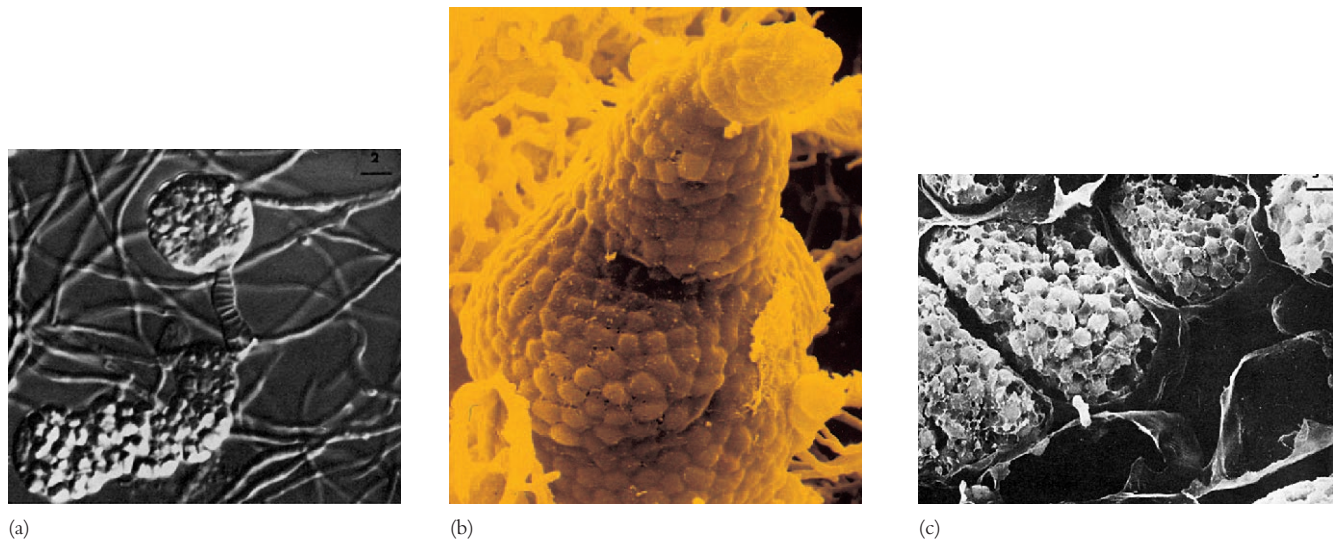
The suborder *Streptosporangineae* contains three families and 14 genera. The genera in this suborder are found in section 30 (maduromycetes) and section 31 (*Thermomonospora* and related genera) of the first edition of *Bergey's Manual*. All genera of the maduromycetes have type III cell walls and the sugar derivative **madurose** (3-*O*-methyl-D-galactose) in whole cell homogenates. Their G + C content is 64 to 74 mol%. Aerial mycelia bear pairs or short chains of spores, and the substrate mycelium is branched (figure 24.2e and figure 24.16). Some genera form sporangia; spores are not heat resistant. Maduromycete genera such as *Streptosporangium*, *Microbispora*, and *Planomonospora* will be placed in the family *Streptosporangiaceae* in the second edition. *Thermomonospora*, *Actinomadura* and *Spirillospora* will be in the family *Thermomonosporaceae*. *Actinomadura* is another actinomycete associated with the disease actinomycetoma. *Thermomonospora* produces single spores on the aerial mycelium or on both the aerial and substrate mycelia. It has been isolated from high-temperature habitats such as compost piles and hay; it can grow at 40 to 48°C.

## 24.10 Suborder Frankineae

Like the suborder *Streptosporangineae*, *Frankineae* contains genera from different parts of the first edition. Two of the genera, *Frankia* and *Geodermatophilus*, come from section 27 (actinomycetes with multilocular sporangia). These genera form clusters of spores when a hypha divides both transversely and longitudinally. (Multilocular means having many cells or compartments.) They have type III cell walls (table 24.1), although the cell extract sugar patterns differ. The G + C content varies from 57 to 75 mol%. *Geodermatophilus* (type IIIC) has motile spores and is an aerobic soil organism. *Frankia* (type IIID) forms nonmotile sporangiospores in a sporogenous body (figure 24.17). It grows in symbiotic association with the roots of at least eight families of higher nonleguminous plants (e.g., alder trees) and is a microaerophile able to fix atmospheric nitrogen (see section 30.4).

The roots of infected plants develop nodules that fix nitrogen so efficiently that a plant such as an alder can grow in the absence of combined nitrogen when nodulated. Within the nodule cells, *Frankia* forms branching hyphae with globular vesicles at their ends (figure 24.17c). These vesicles may be the sites of nitrogen fixation. The nitrogen-fixation process resembles that of *Rhizobium* in that it is oxygen sensitive and requires molybdenum and cobalt.



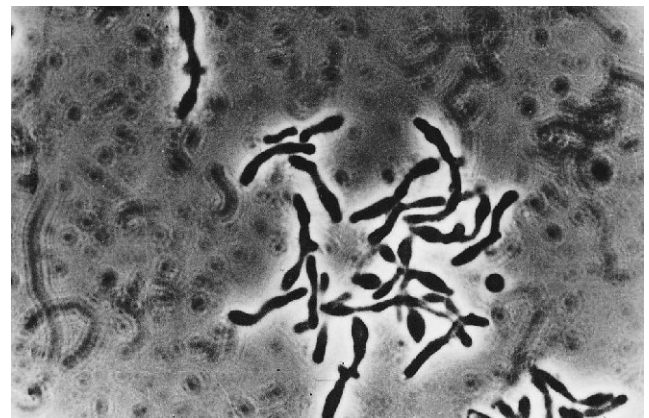


**Figure 24.17** *Frankia*. (a) An interference contrast micrograph showing hyphae, multilocular sporangia, and spores. Bar = 5  $\mu$ m. (b) A scanning electron micrograph of a sporangium surrounded by hyphae. (c) A nodule of the alder *Alnus rubra* showing cells filled with vesicles of *Frankia*. Scanning electron microscopy. Bar = 5  $\mu$ m.

Another genus in this suborder, *Sporichthya*, has been transferred from section 29 (streptomycetes and related genera) of the first edition. *Sporichthya* is one of the strangest of the actinomycetes. It lacks a substrate mycelium. The hyphae remain attached to the substratum by holdfasts and grow upward to form aerial mycelia that release motile, flagellate conidia in the presence of water.

## 24.11 Order *Bifidobacteriales*

The order *Bifidobacteriales* has one true family, *Bifidobacteriaceae*, and eight genera (five of which have unknown affiliation). *Falcivibrio* and *Gardnerella* are found in the human genital/urinary tract; *Gardnerella* is thought to be a major cause of bacterial vaginitis. *Bifidobacterium* probably is the best-studied genus. Bifidobacteria are nonmotile, nonsporing, gram-positive rods of varied shapes that are slightly curved and clubbed; often they are branched (**figure 24.18**). The rods can be single or in clusters and V-shaped pairs. *Bifidobacterium* is anaerobic and actively ferments carbohydrates to produce acetic and lactic acids, but no carbon dioxide. It is found in the mouth and intestinal tract of warm-blooded vertebrates, in sewage, and in insects. *B. bifidus* is a pioneer colonizer of the human intestinal tract, particularly when babies are breast fed. A few *Bifidobacterium* infections have been reported in humans, but the genus does not appear to be a major cause of disease.



**Figure 24.18** *Bifidobacterium*. *Bifidobacterium bifidum*; phase-contrast photomicrograph ( $\times 1,500$ ).

1. Give the distinguishing properties of the actinoplanetes.
2. Briefly describe the variations in sporangia and sporophore organization of the actinoplanetes.
3. Describe the genus *Propionibacterium* and comment on its practical importance.

4. What characteristics do streptomycetes have in common?
5. Describe the major properties of the genus *Streptomyces*.
6. Give three ways in which *Streptomyces* is of ecological and medical importance.
7. Briefly describe the genera of the suborder *Streptosporangineae*. What is madurose? Why is *Actinomadura* important?
8. Describe *Frankia* and discuss its importance.
9. Characterize the genus *Bifidobacterium*. Where is it found and why is it significant?



### Summary

1. Actinomycetes are aerobic, gram-positive bacteria that form branching, usually nonfragmenting, hyphae and asexual spores (figure 24.1).
2. The asexual spores borne on aerial mycelia are called conidiospores or conidia if they are at the tip of hyphae and sporangiospores when they are within sporangia.
3. Actinomycetes have several distinctively different types of cell walls and often also vary in terms of the sugars present in cell extracts. Properties such as color and morphology are also taxonomically useful.
4. The first edition of *Bergey's Manual* classifies the actinomycetes based on such properties as conidial arrangement, the presence of a sporangium, cell wall type, and cell extract sugars (tables 24.1–24.3).
5. The second edition of *Bergey's Manual* classifies the high G + C bacteria phylogenetically using 16S rRNA data. The phylum *Actinobacteria* contains one class, five subclasses, six orders, 14 suborders, and 40 families (figure 24.3).
6. The suborder *Actinomycineae* contains the genus *Actinomyces*, members of which are irregularly shaped, nonsporulating rods that can cause disease in cattle and humans.
7. The suborder *Micrococcineae* has the genera *Micrococcus*, *Arthrobacter*, and *Dermatophilus*. *Arthrobacter* has an unusual rod-coccus growth cycle and carries out snapping division (figure 24.7).
8. The genera *Corynebacterium*, *Mycobacterium*, and *Nocardia* will be placed in the suborder *Corynebacterineae*. *Mycobacteria* form either rods or filaments that readily fragment. Their cell walls have a high lipid content and mycolic acids; the presence of these lipids makes them acid-fast. The genera *Corynebacterium* and *Mycobacterium* contain several very important human pathogens.
9. Nocardioform actinomycetes have hyphae that readily fragment into rods and coccoid elements, and often form aerial mycelia with spores.
10. The suborder *Micromonosporineae* has genera that are classified as actinoplanetes in section 28 of the first edition. These actinomycetes have an extensive substrate mycelium and form special aerial sporangia (figure 24.12). They are present in soil, fresh water, and the ocean. The soil forms are probably important in decomposition.
11. The genus *Propionibacterium* has been placed in the suborder *Propionibacterineae* in the second edition. Members of this genus are common skin and intestinal inhabitants and are important in cheese manufacture and the development of acne vulgaris.
12. The suborder *Streptomycineae* of the second edition contain the genus *Streptomyces*. Members of this genus have type I cell walls and aerial hyphae bearing chains of 3 to 50 or more nonmotile conidiospores within a thin sheath (figure 24.13).
13. Streptomycetes are important in the degradation of more resistant organic material in the soil and produce many useful antibiotics. A few cause diseases in plants and animals.
14. The suborder *Streptosporangineae* contains several genera that the first edition places in section 30 (the maduromycetes) and section 31 (*Thermomonospora* and related genera). Many of these organisms have the sugar derivative madurose and type III cell walls.
15. The genera *Frankia* and *Geodermatophilus* are placed in the suborder *Frankineae*. They produce clusters of spores at hyphal tips and have type III cell walls. *Frankia* grows in symbiotic association with higher nonleguminous plants and fixes nitrogen.
16. The genus *Bifidobacterium* is placed in the order *Bifidobacteriales* in the second edition. This irregular, anaerobic rod is one of the first colonizers of the intestinal tract in nursing babies.

### Key Terms

acid-fast 543  
actinobacteria 541  
actinomycete 537  
actinomycetoma 547  
conidia 537

conidiospores 537  
geosmin 546  
madurose 548  
mycolic acids 543  
nocardioforms 544

snapping division 542  
sporangiospores 539  
streptomycetes 546  
thallus 537

### Questions for Thought and Review

1. Suppose that you discovered a nodulated plant that could fix atmospheric nitrogen. How might you show that a bacterial symbiont was involved and that *Frankia* rather than *Rhizobium* was responsible?
2. How could one decide whether a newly isolated spore-forming bacterium was producing conidiospores or true endospores?
3. List the properties that are most useful in actinomycete taxonomy and give some indication of their relative importance. How does the second edition of *Bergey's Manual* differ in its approach to classification of actinomycetes?
4. Which bacteria are noted for each of the following: the rod-coccus growth cycle, early colonization of the human intestine, metachromatic granules and snapping division, 60 to 90 carbon mycolic acids and acid fastness, the synthesis of antibiotics, the production of substrate mycelia that readily break into rods and cocci, and the production of large amounts of propionic and acetic acids.
5. Propose an explanation for the observation that mycobacteria are normally only weakly gram positive.
6. Which characteristics might be most important in distinguishing between the genera *Corynebacterium*, *Arthrobacter*, *Actinomyces*, *Propionibacterium*, *Mycobacterium*, and *Bifidobacterium*?

### Critical Thinking Questions

1. Even though these are “high G + C” organisms, there are regions of the genome that must be more AT-rich. Suggest a few such regions and explain why they must be more AT-rich.
2. Choose two different species in the phylum *Actinobacteria* and investigate their physiology and ecology. Compare and contrast these two organisms. Can you determine why having a high G + C genomic content might confer an evolutionary advantage?
3. *Streptomyces coelicolor* has a developmental program for differentiation. Some of the genes involved in sporulation contain a rare codon, not used in vegetative genes. Suggest how *Streptomyces* might turn on sporulation, in response to starvation.

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## CHAPTER 25

# The Fungi (Eumycota), Slime Molds, and Water Molds



This is a scanning electron micrograph of the microscopic, unicellular yeast, *Saccharomyces cerevisiae* ( $\times 21,000$ ). *S. cerevisiae* is the most thoroughly investigated eucaryotic microorganism in the world. This has led to a better understanding of the biology of the eucaryotic cell. Today it serves as a widely used biotechnological production organism as well as a eucaryotic model system.

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  - Division Ascomycota 560
  - Division Basidiomycota 561
  - Division Deuteromycota 564
  - Division Chytridiomycota 564
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  - Division Oomycota 565

### Concepts

1. Fungi are widely distributed and are found wherever moisture is present. They are of great importance to humans in both beneficial and harmful ways.
2. Fungi exist primarily as filamentous hyphae. A mass of hyphae is called a mycelium.
3. Like some bacteria, fungi digest insoluble organic matter by secreting exoenzymes, then absorbing the solubilized nutrients.
4. Two reproductive structures occur in the fungi: (1) sporangia form asexual spores, and (2) gametangia form sexual gametes.
5. The zygomycetes are characterized by resting structures called zygospores—cells in which zygotes are formed.
6. The ascomycetes form zygotes within a characteristic saclike structure, the ascus. The ascus contains two or more ascospores.
7. Yeasts are unicellular fungi—mainly ascomycetes.
8. Basidiomycetes possess dikaryotic hyphae with two nuclei, one of each mating type. The hyphae divide uniquely, forming basidiocarps within which club-shaped basidia can be found. The basidia bear two or more basidiospores.



## Concepts (continued)

9. The deuteromycetes (Fungi Imperfecti) have either lost the capacity for sexual reproduction, or it has never been observed.
10. The chytrids are a group of terrestrial and aquatic fungi that reproduce by motile zoospores with single, posterior, whiplash flagella, and represent a link between the true fungi and protists.
11. The slime and water molds resemble the fungi only in appearance and life-style. In their cellular organization, reproduction, and life cycles, they are phylogenetically distinct.

*Yeasts, molds, mushrooms, mildews, and the other fungi pervade our world. They work great good and terrible evil. Upon them, indeed, hangs the balance of life; for without their presence in the cycle of decay and regeneration, neither man nor any other living thing could survive.*

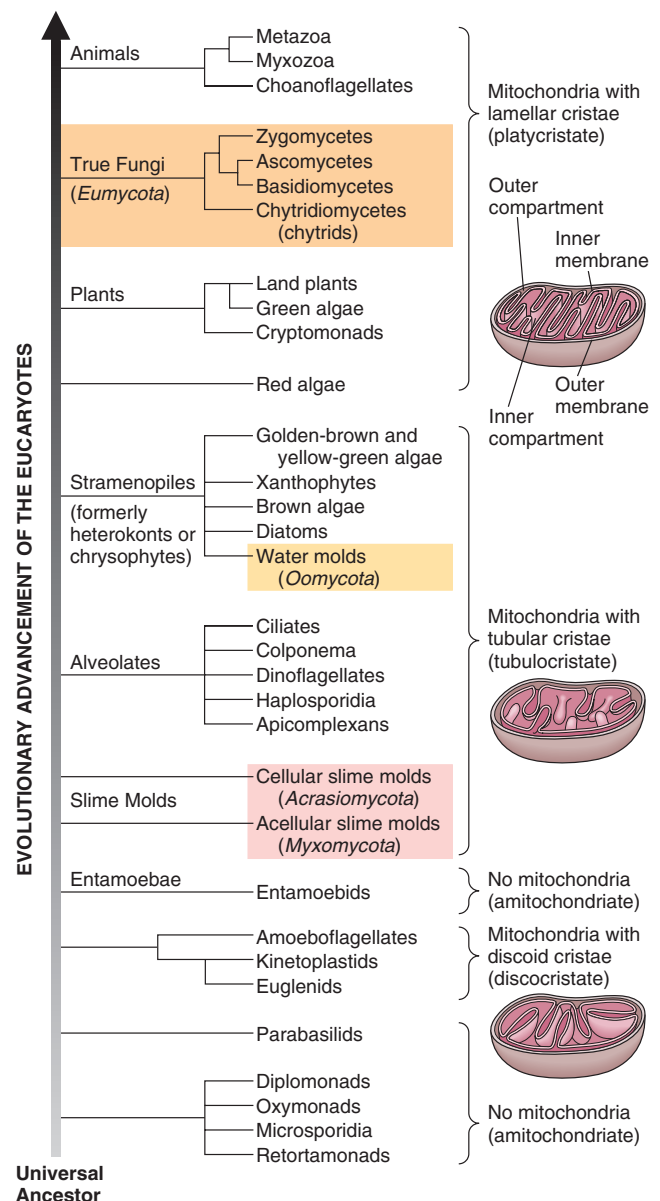
—Lucy Kavaler

This chapter is an introduction to the true fungi. It surveys their diversity, discusses their ecological and commercial impact, and presents some of their typical life cycles. A brief overview is also given of the eucaryotes that resemble the fungi—the slime molds and water molds.

Microbiologists use the term **fungus** [pl., fungi; Latin *fungus*, mushroom] to include eucaryotic, spore-bearing organisms with absorptive nutrition, no chlorophyll, and that reproduce sexually and asexually. Scientists who study fungi are **mycologists** [Greek *mykes*, mushroom, and *logos*, discourse], and the scientific discipline dealing with fungi is called **mycology**. The study of fungal

**Phylogenetic Diagram 25** Tentative Phylogeny of the True Fungi (*Eumycota*), Slime Molds, and Water Molds Based on 18S rRNA Sequence Comparisons. Using molecular systematics, organisms are grouped together based on the molecular phylogeny of their nuclear SSU rRNA genes and the type of mitochondrial cristae present. According to this tentative phylogeny, the true fungi, slime molds, and water molds form three distinct monophyletic branches (highlighted by three different colors). Accordingly, the four major groups of fungi (Zygomycetes, Ascomycetes, Basidiomycetes, and Chytridiomycetes) form a phylogenetically coherent group. Molecular characters have been essential for phylogenetic analysis in cases when morphological characters are convergent, reduced, or missing. This is especially true for species that never reproduce sexually (e.g., the Fungi Imperfecti or *Deuteromycota*), because characters of sexual reproduction traditionally have been the basis for classifying fungi. The use of molecular characters allows the asexual fungi (*Deuteromycota*) to be placed among their closest relatives in the *Eumycota* and not in a separate division.

toxins and their effects is called **mycotoxicology**, and the diseases caused by fungi in animals are known as **mycoses** [s., mycosis]. The five-kingdom system places the fungi in the kingdom *Fungi* (see figure 19.12a). According to the universal phylogenetic tree, fungi are members of the domain *Eucarya* (see figure 19.3). As presently delimited, the kingdom *Fungi* is believed to constitute a monophyletic group (**Phylogenetic Diagram 25**) referred to as the true fungi or *Eumycota* (Greek *eu*, true, and *mykes*, fungus).



## 25.1 Distribution

Fungi are primarily terrestrial organisms, although a few are freshwater or marine. Many are pathogenic and infect plants and animals. Fungi also form beneficial relationships with other organisms. For example, about three-fourths of all vascular plants form associations (called mycorrhizae) between their roots and fungi. Fungi also are found in the upper portions of many plants. These endophytic fungi affect plant reproduction and palatability to herbivores. Lichens are associations of fungi and either algae or cyanobacteria. [Mycorrhizae](#) (pp. 679–82); [Endophytic fungi](#) (p. 679); [Lichens](#) (pp. 598–99)

## 25.2 Importance

About 90,000 fungal species have been described; however, some estimates of total numbers suggest that 1.5 million species may exist.

Fungi are important to humans in both beneficial and harmful ways. With bacteria and a few other groups of heterotrophic organisms, fungi act as decomposers, a role of enormous significance. They degrade complex organic materials in the environment to simple organic compounds and inorganic molecules. In this way carbon, nitrogen, phosphorus, and other critical constituents of dead organisms are released and made available for living organisms (*see section 28.3*).

However, fungi are the major cause of plant diseases (**figure 25.1a**). Over 5,000 species attack economically valuable crops and garden plants and also many wild plants. In like manner many diseases of animals (**figure 25.1b** and **table 25.1**) and humans (*see section 40.1*) are caused by fungi.

Fungi, especially the yeasts, are essential to many industrial processes involving fermentation (*see chapter 41*). Examples include the making of bread, wine, and beer. Fungi also play a major role in the preparation of some cheeses, soy sauce, and sufu; in the commercial production of many organic acids (citric, gallic) and certain drugs (ergometrine, cortisone); and in the manufacture of many antibiotics (penicillin, griseofulvin) and the immunosuppressive drug cyclosporine.

In addition, fungi are important research tools in the study of fundamental biological processes. Cytologists, geneticists, biochemists, biophysicists, and microbiologists regularly use fungi in their research. Based on this research the yeast *Saccharomyces cerevisiae* is the best understood eucaryotic cell.

## 25.3 Structure

The body or vegetative structure of a fungus is called a **thallus** [pl., thalli]. It varies in complexity and size, ranging from the single-cell microscopic yeasts to multicellular molds, macroscopic



(a)



(b)

**Figure 25.1 Fungal Diseases.** (a) Apple scab. Midsummer scab lesions on fruit and leaf, that developed from spring infections. These lesions produce spores capable of causing further spread of the disease. (b) Dermatomycosis. Ringworm in a kitten due to *Microsporum canis*.

puffballs, and mushrooms (**figure 25.2**). The fungal cell usually is encased in a cell wall of **chitin**. Chitin is a strong but flexible nitrogen-containing polysaccharide consisting of *N*-acetylglucosamine residues.

A **yeast** is a unicellular fungus that has a single nucleus and reproduces either asexually by budding and transverse division or sexually through spore formation. Each bud that separates can grow into a new yeast, and some group together to form colonies. Generally yeast cells are larger than bacteria, vary considerably in size, and are commonly spherical to egg shaped. They have no flagella but do possess most of the other eucaryotic organelles (**figure 25.3**).

**Table 25.1** Some Mycotoxicoses<sup>a</sup> Produced by Fungal Mycotoxins in Domestic Animals

| Disease                      | Fungus                                                | Mycotoxin           | Contaminated Foodstuff                          | Animals Affected                    |
|------------------------------|-------------------------------------------------------|---------------------|-------------------------------------------------|-------------------------------------|
| Aflatoxicosis                | <i>Aspergillus flavus</i>                             | Aflatoxins          | Rice, corn, sorghum, cereals, peanuts, soybeans | Poultry, swine, cattle, sheep, dogs |
| Ergotism                     | <i>Claviceps purpurea</i>                             | Ergot alkaloids     | Seedheads of many grasses, grains               | Cattle, horses, swine, poultry      |
| Mushroom poisoning           | <i>Amanita verna</i>                                  | Amanitins           | Eaten from pastures                             | Cattle                              |
| Poultry hemorrhagic syndrome | <i>Aspergillus flavus</i> and others                  | Aflatoxins          | Toxic grain and meal                            | Chickens                            |
| Slobbers                     | <i>Rhizoctonia</i>                                    | Alkaloid slaframine | Red clover                                      | Sheep, cattle                       |
| Tall fescue toxicosis        | <i>Acremonium coenophialum</i> (an endophytic fungus) | Ergot alkaloids     | Endophyte-infected tall fescue plants           | Cattle, horses                      |

<sup>a</sup>A mycotoxicosis [pl., mycotoxicoses] is a poisoning caused by a fungal toxin.



(a)

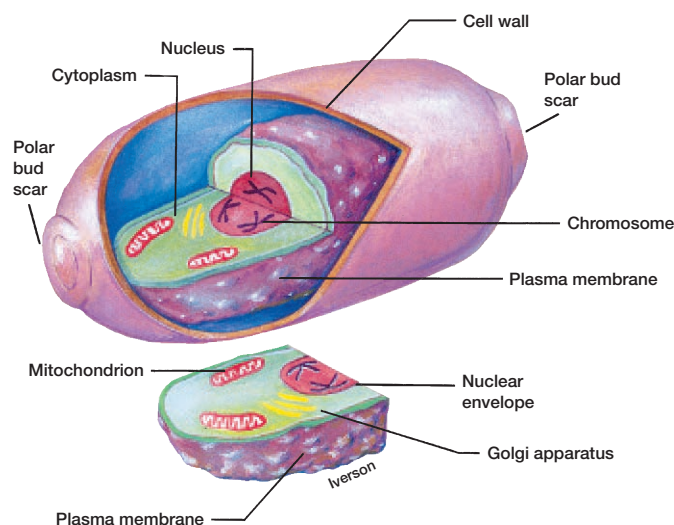


(b)



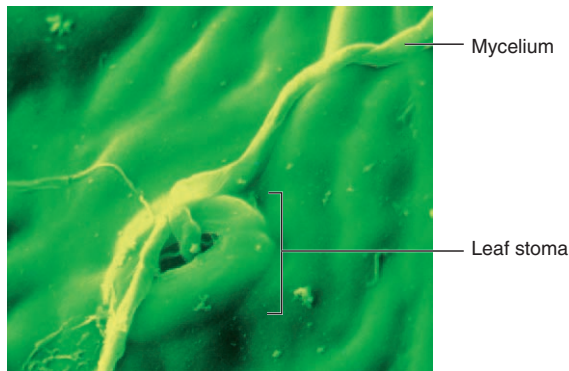
(c)

**Figure 25.2 Fungal Thalli.** (a) The multicellular common mold, *Penicillium*, growing on an apple. (b) A large group of puffballs, *Lycoperdon*, growing on a log. (c) A mushroom is made up of densely packed hyphae that form the mycelium or visible structure (thallus).



**Figure 25.3 A Yeast.** Diagrammatic drawing of a yeast cell showing typical morphology. For clarity, the plasma membrane has been drawn separated from the cell wall. In a living cell the plasma membrane adheres tightly to the cell wall.



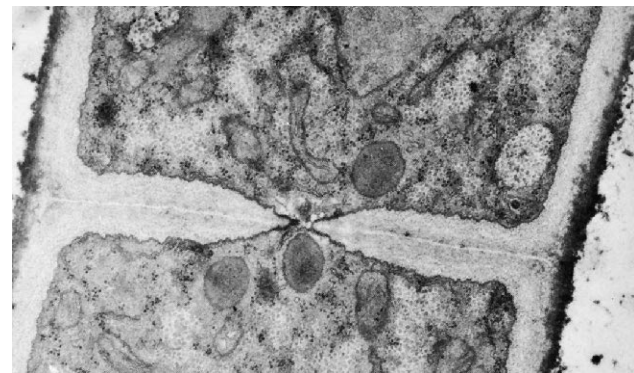
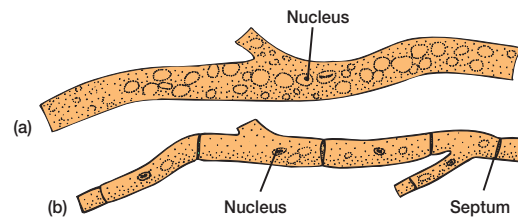


(a)

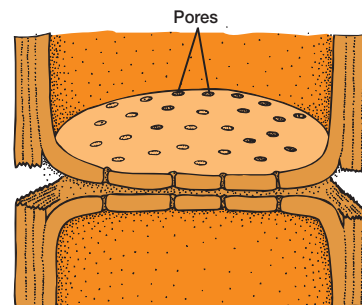


(b)

**Figure 25.4 Mold Mycelia.** (a) Scanning electron micrograph of a young mycelial aggregate forming over a leaf stoma ( $\times 1,000$ ). (b) A very large macroscopic mycelium of a basidiomycete growing on the soil.



(c)



(d)

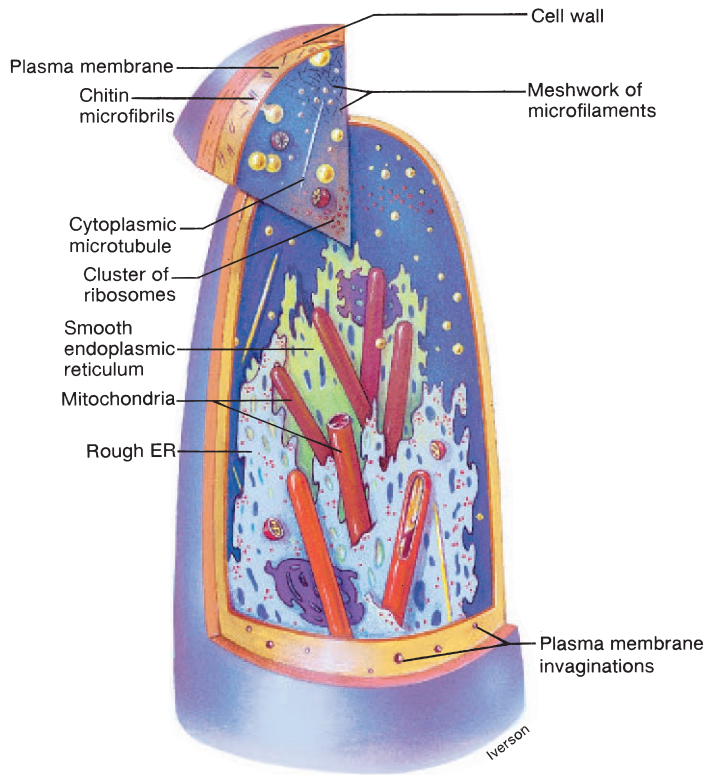
**Figure 25.5 Hyphae.** Drawings of (a) coenocytic hyphae and (b) hyphae divided into cells by septa. (c) Electron micrograph ( $\times 40,000$ ) of a section of *Drechslera sorokiniana* showing wall differentiation and a single pore. (d) Drawing of a multiperforate septal wall structure.

A **mold** (figure 25.2a) consists of long, branched, threadlike filaments of cells called **hyphae** [s., hypha; Greek *hyphe*, web] that form a **mycelium** [pl., mycelia], a tangled mass or tissuelike aggregation (figure 25.4). In some fungi, protoplasm streams through hyphae, uninterrupted by cross walls. These hyphae are called **coenocytic** (figure 25.5a). The hyphae of other fungi (figure 25.5b) have cross walls called **septa** [s., septum] with either a single pore (figure 25.5c) or multiple pores (figure 25.5d) that permit cytoplasmic streaming. These hyphae are termed **septate**.

Hyphae are composed of an outer cell wall and an inner lumen, which contains the cytosol and organelles (figure 25.6). A

plasma membrane surrounds the cytoplasm and lies next to the cell wall.

Many fungi, especially those that cause diseases in humans and animals, are dimorphic (table 25.2)—that is, they have two forms. Dimorphic fungi can change from (1) the yeast (Y) form in the animal to (2) the mold or mycelial form (M) in the external environment in response to changes in various environmental factors (nutrients, CO<sub>2</sub> tension, oxidation-reduction potentials, temperature). This shift is called the **YM shift**. In plant-associated fungi the opposite type of dimorphism exists: the mycelial form occurs in the plant and the yeast form in the external environment.



**Figure 25.6 Hyphal Morphology.** Diagrammatic representation of hyphal tip showing typical organelles and other structures.

**Table 25.2** Some Medically Important Dimorphic Fungi

| Fungus                               | Disease <sup>a</sup>   |
|--------------------------------------|------------------------|
| <i>Blastomyces dermatitidis</i>      | Blastomycosis          |
| <i>Candida albicans</i>              | Candidiasis            |
| <i>Coccidioides capsulatum</i>       | Coccidioidomycosis     |
| <i>Histoplasma capsulatum</i>        | Histoplasmosis         |
| <i>Sporothrix schenckii</i>          | Sporotrichosis         |
| <i>Paracoccidioides brasiliensis</i> | Paracoccidioidomycosis |

<sup>a</sup>See chapter 40 for a discussion of each of these diseases.

## 25.4 Nutrition and Metabolism

Fungi grow best in dark, moist habitats, but they are found wherever organic material is available. Most fungi are **saprophytes**, securing their nutrients from dead organic material. Like many bacteria, fungi release hydrolytic exoenzymes that digest external substrates. They then absorb the soluble products. They are chemoorganoheterotrophs and use organic compounds as a source of carbon, electrons, and energy.

Glycogen is the primary storage polysaccharide in fungi. Most fungi use carbohydrates (preferably glucose or maltose) and nitrogenous compounds to synthesize their own amino acids and proteins.

Fungi usually are aerobic. Some yeasts, however, are facultatively anaerobic and can obtain energy by fermentation, such as in the production of ethyl alcohol from glucose. Obligately anaerobic fungi are found in the rumen of cattle.

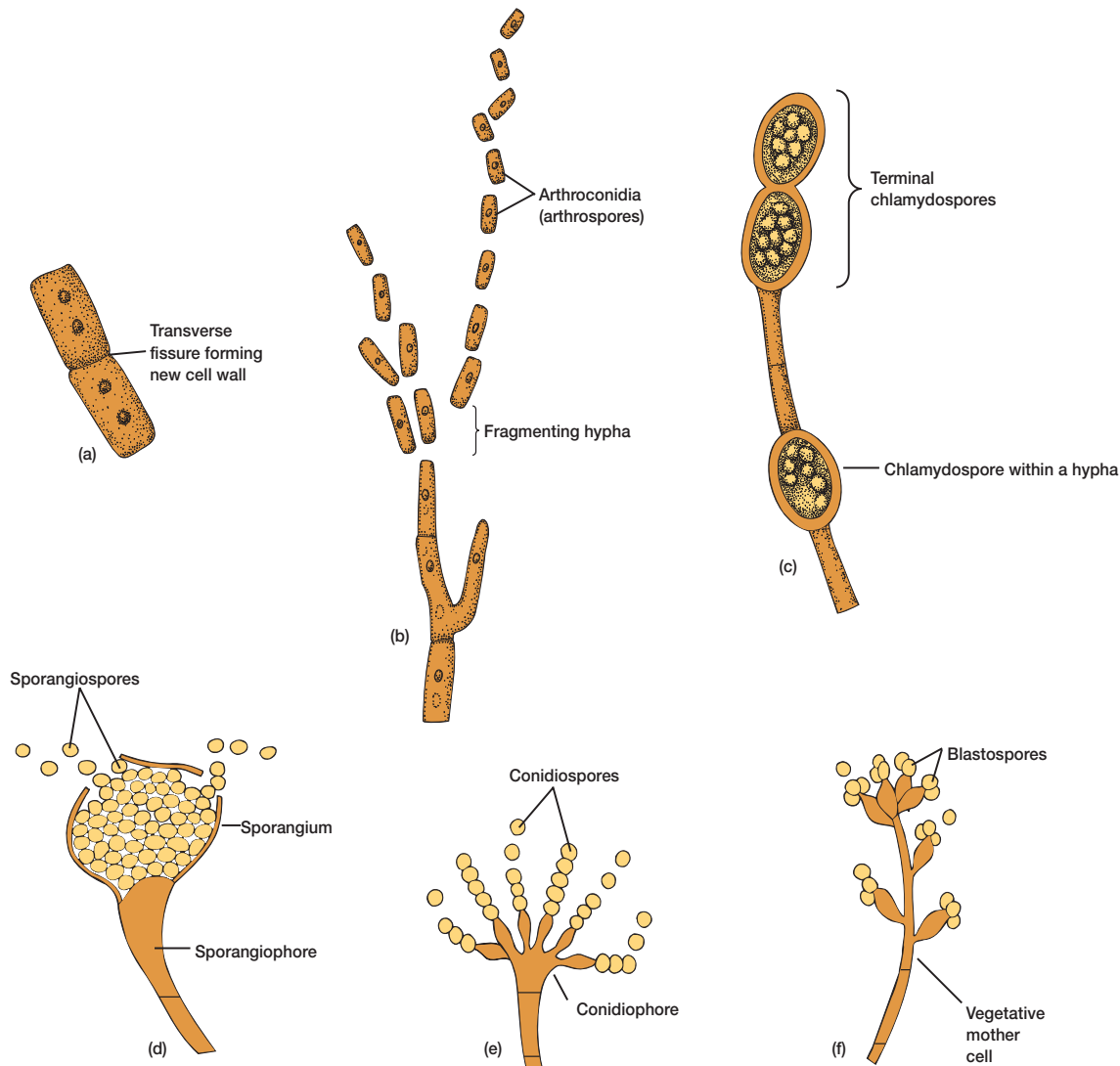
## 25.5 Reproduction

Reproduction in fungi can be either asexual or sexual. Asexual reproduction is accomplished in several ways:

1. A parent cell can divide into two daughter cells by central constriction and formation of a new cell wall (**figure 25.7a**).
2. Somatic vegetative cells may bud to produce new organisms. This is very common in the yeasts.
3. The most common method of asexual reproduction is spore production. Asexual spore formation occurs in an individual fungus through mitosis and subsequent cell division. There are several types of asexual spores:
  - a. A hypha can fragment (by the separation of hyphae through splitting of the cell wall or septum) to form cells that behave as spores. These cells are called **arthroconidia** or **arthrospores** (**figure 25.7b**).
  - b. If the cells are surrounded by a thick wall before separation, they are called **chlamydospores** (**figure 25.7c**).
  - c. If the spores develop within a sac [**sporangium**; pl., **sporangia**] at a hyphal tip, they are called **sporangiospores** (**figure 25.7d**).
  - d. If the spores are not enclosed in a sac but produced at the tips or sides of the hypha, they are termed **conidiospores** (**figures 25.7e and 25.11**).
  - e. Spores produced from a vegetative mother cell by budding (**figure 25.7f**) are called **blastospores**.

Sexual reproduction in fungi involves the union of compatible nuclei. Some fungal species are self-fertilizing and produce sexually compatible gametes on the same mycelium (homothallic). Other species require outcrossing between different but sexually compatible mycelia (heterothallic). Depending on the species, sexual fusion may occur between haploid gametes, gamete-producing bodies called **gametangia**, or hyphae. Sometimes both the cytoplasm and haploid nuclei fuse immediately to produce the diploid zygote. Usually, however, there is a delay between cytoplasmic and nuclear fusion. This produces a **dikaryotic stage** in which cells contain two

1. How can a fungus be defined?
2. Where are fungi found?
3. Why are fungi important as decomposers?
4. What are some forms represented by different fungal thalli?
5. What organelles would you expect to find in the cytoplasm of a typical fungus?
6. Describe a typical yeast; a typical mold.



**Figure 25.7** Diagrammatic Representation of Asexual Reproduction in the Fungi and Some Representative Spores. (a) Transverse fission. (b) Hyphal fragmentation resulting in arthroconidia (arthrospores) and (c) chlamydospores. (d) Sporangiospores in a sporangium. (e) Conidiospores arranged in chains at the end of a conidiophore. (f) Blastospores are formed from buds off of the parent cell.

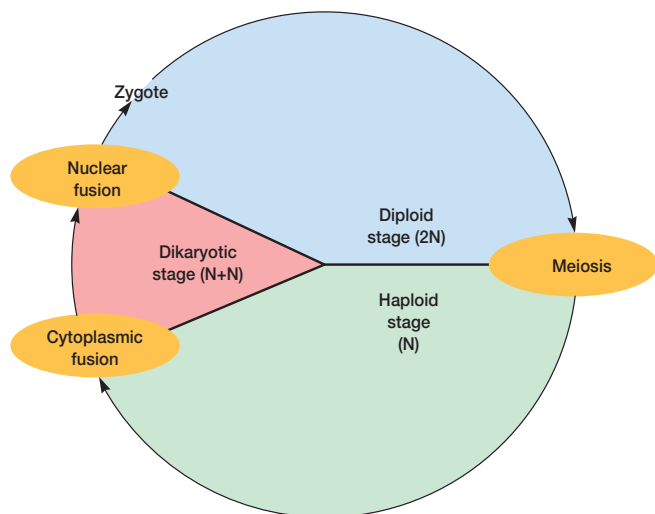
separate haploid nuclei, one from each parent (**figure 25.8**). After a period of dikaryotic existence, the two nuclei fuse. This sexual reproduction yields spores. For example, in the zygomycetes the zygote develops into a **zygospore** (**figure 25.9**); in the ascomycetes, an **ascospore** (**figure 25.12**); and in the basidiomycetes, a **basidiospore** (**figure 25.14**).

Fungal spores are important for several reasons. The size, shape, color, and number of spores are useful in the identification of fungal species. The spores are often small and light; they can remain suspended in air for long periods. Thus they frequently aid in fungal dissemination, a significant factor that explains the wide

distribution of many fungi. Fungal spores often spread by adhering to the bodies of insects and other animals. The bright colors and fluffy textures of many molds often are due to their aerial hyphae and spores.

1. What generally governs the ecological distribution of fungi?
2. How is asexual reproduction accomplished in the fungi? Sexual reproduction?
3. Describe each of the following types of asexual fungal spores: sporangiospore, conidiospore, and blastospore.





**Figure 25.8 Reproduction in Fungi.** A drawing of the generalized life cycle for fungi showing the alternation of haploid and diploid stages. Some fungal species do not pass through the dikaryotic stage indicated in this drawing. The asexual (haploid) stage is used to produce spores that aid in the dissemination of the species. The sexual (diploid) stage involves the formation of spores that survive adverse environmental conditions (e.g., cold, dryness, heat).

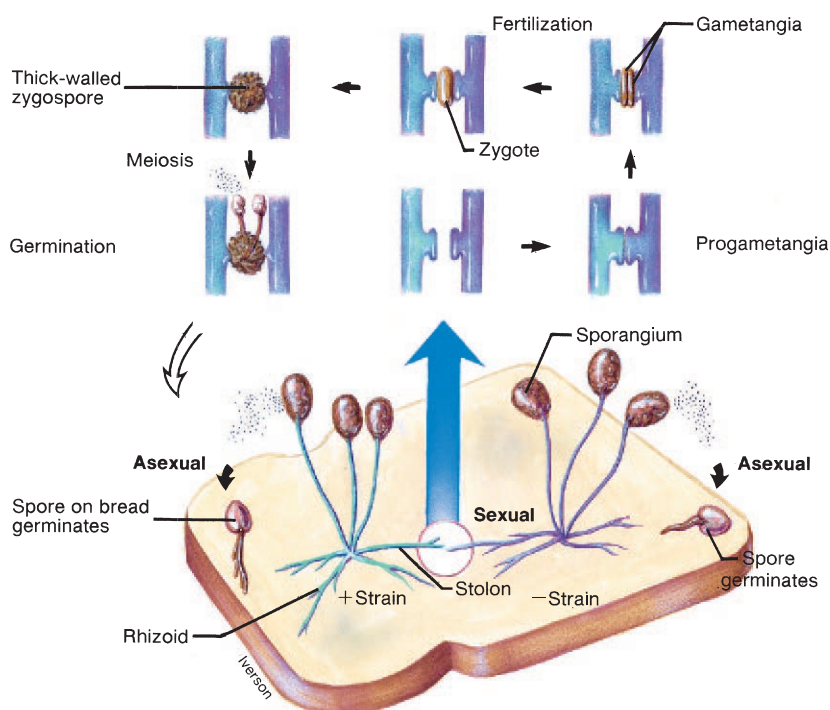
## 25.6 Characteristics of the Fungal Divisions

The traditional taxonomic scheme used by mycologists classifies the fungi into four divisions (**table 25.3**), based primarily on variations in sexual reproduction. (In mycology a division is equivalent to a phylum in animal classification schemes.) Based on 18S rRNA studies, molecular microbiologists place the *Deuteromycota* (Fungi Imperfecti) among their closest relatives in either the *Zygomycota*, *Ascomycota*, or *Basidiomycota* and add the class *Chytridiomycetes* (Phylogenetic Diagram 25).

**Table 25.3 Divisions of Fungi\***

| Division                          | Common Name      | Approximate Number of Species |
|-----------------------------------|------------------|-------------------------------|
| <i>Zygomycota</i>                 | Zygomycetes      | 600                           |
| <i>Ascomycota</i>                 | Sac fungi        | 35,000                        |
| <i>Basidiomycota</i>              | Club fungi       | 30,000                        |
| <i>Deuteromycota</i> <sup>a</sup> | Fungi Imperfecti | 30,000                        |

<sup>a</sup>According to the traditional system often followed by mycologists.



**Figure 25.9 Division Zygomycota.** Diagrammatic representation of the life cycle of *Rhizopus stolonifer*. Both the sexual and asexual phases are illustrated.



**Figure 25.10** Division *Ascomycota*. (a) The common morel, *Morchella esculenta*, is one of the choicest edible fungi. It fruits in the spring. (b) Scarlet cups, *Sarcoscypha coccinea*, with open ascocarps (apothecia). (c) The black truffle, *Tuber brumale*, is highly prized for its flavor by gourmet cooks. Technically truffles are mycorrhizal associations on oak trees.

### Division *Zygomycota*

The division *Zygomycota* contains the fungi called **zygomycetes**. Most live on decaying plant and animal matter in the soil; a few are parasites of plants, insects, animals, and humans. The hyphae of zygomycetes are coenocytic, with many haploid nuclei. Asexual spores, usually wind dispersed, develop in sporangia at the tips of aerial hyphae. Sexual reproduction produces tough, thick-walled zygotes called zygospores that can remain dormant when the environment is too harsh for growth of the fungus.

The bread mold, *Rhizopus stolonifer*, is a very common member of this division. This fungus grows on the surface of moist, carbohydrate-rich foods, such as breads, fruits, and vegetables. On breads, for example, *Rhizopus*'s hyphae rapidly cover the surface. Special hyphae called rhizoids extend into the bread, and absorb nutrients (figure 25.9). Other hyphae (stolons) become erect, then arch back into the substratum forming new rhizoids. Still others remain erect and produce at their tips asexual sporangia filled with the black spores, giving the mold its characteristic color. Each spore, when liberated, can start a new mycelium.

*Rhizopus* usually reproduces asexually, but if food becomes scarce or environmental conditions unfavorable, it begins sexual reproduction. Sexual reproduction requires compatible strains of opposite mating types (figure 25.9). These have traditionally been labeled + and – strains because they are not morphologically distinguishable as male and female. When the two mating strains are close, hormones are produced that cause their hyphae to form

projections called **progametangia** [Greek *pro*, before], and then mature gametangia. After fusion of the gametangia, the nuclei of the two gametes fuse, forming a zygote. The zygote develops a thick, rough, black coat and becomes a dormant zygospore. Meiosis often occurs at the time of germination; the zygospore then splits open and produces a hypha that bears an asexual sporangium and the cycle begins anew.

The zygomycetes also contribute to human welfare. For example, *Rhizopus* is used in Indonesia to produce a food called tempeh from boiled, skinless soybeans. Another zygomycete (*Mucor* spp.) is used with soybeans in the Orient to make a cheese called sufu. Others are employed in the commercial preparation of some anesthetics, birth control agents, industrial alcohols, meat tenderizers, and the yellow coloring used in margarine and butter substitutes.

### Division *Ascomycota*

The division *Ascomycota* contains the fungi called **ascomycetes**, commonly known as the sac fungi. Many species are quite familiar and economically important (figure 25.10). For example, most of the red, brown, and blue-green molds that cause food spoilage are ascomycetes. The powdery mildews that attack plant leaves and the fungi that cause chestnut blight and Dutch elm disease are ascomycetes. Many yeasts as well as edible morels and truffles are ascomycetes. The pink bread mold *Neurospora crassa*, also an ascomycete, has been a most important research tool in genetics and biochemistry.



**Figure 25.11** Asexual Reproduction in Ascomycetes.  
Characteristic conidiospores of *Aspergillus* as viewed with the electron microscope ( $\times 1,200$ ).

Many ascomycetes are parasites on higher plants. *Claviceps purpurea* parasitizes rye and other grasses, causing the disease **ergot**. **Ergotism**, the toxic condition in humans and animals who eat grain infected with the fungus, is often accompanied by gangrene, psychotic delusions, nervous spasms, abortion, and convulsions. During the Middle Ages ergotism, then known as St. Anthony's fire, killed thousands of people. For example, over 40,000 deaths from ergot poisoning were recorded in France in the year 943. It has been suggested that the widespread accusations of witchcraft in Salem Village (now Danvers) and other Massachusetts communities in the late 1690s may have resulted from outbreaks of ergotism. The pharmacological activities of ergot are due to its active ingredient, lysergic acid diethylamide (LSD). In controlled dosages ergot can be used to induce labor, lower blood pressure, and ease migraine headaches.

The ascomycetes are named for their characteristic reproductive structure, the saclike **ascus** [pl., **asci**; Greek *askos*, sac]. The mycelium of the ascomycetes is composed of septate hyphae. Asexual reproduction is common in the ascomycetes and takes place by way of conidiospores (**figure 25.11**).

Sexual reproduction in the ascomycetes always involves the formation of an ascus containing two or more haploid ascospores (**figure 25.12a**). In the more complex ascomycetes, ascus formation is preceded by the development of special **ascogamous hyphae** into which pairs of nuclei migrate (**figure 25.12b**). One nucleus of each pair originates from a "male" mycelium (**antheridium**) or cell and the other from a "female" organ or cell (**ascogonium**) that has fused with it. As the ascogamous hyphae grow, the paired nuclei divide so that there is one pair of nuclei in each cell. After the ascogamous hyphae have matured, nuclear fusion occurs at the hyphal tips in the as-

cus mother cells. The diploid zygote nucleus then undergoes meiosis, and the resulting four haploid nuclei divide mitotically again to produce a row of eight nuclei in each developing ascus. These nuclei are walled off from one another. Thousands of asci may be packed together in a cup- or flask-shaped **ascocarp** (**figure 25.10b**). When the ascospores mature, they often are released from the asci with great force. If, perchance, the mature ascocarp is jarred, it may appear to belch puffs of "smoke" consisting of thousands of ascospores. Upon reaching a suitable environment, the ascospores germinate and start the cycle anew.

Although the term yeast is used in a general sense to refer to all unicellular fungi that reproduce asexually by either budding or binary fission (**figure 25.13a,b**), many yeast genera are classified specifically within the ascomycetes because of their sexual reproduction (**figure 25.13c,d**). Yeasts are present in both terrestrial and aquatic habitats in which a suitable carbon source is available.

### Division Basidiomycota

The division *Basidiomycota* contains the **basidiomycetes**, commonly known as the club fungi. Examples include the smuts, jelly fungi, rusts, shelf fungi, stinkhorns, puffballs, toadstools, mushrooms, and bird's nest fungi.

Basidiomycetes are named for their characteristic structure or cell, the **basidium**, that is involved in sexual reproduction (**figure 25.14**). A basidium [Greek *basidion*, small base] is produced at the tip of hyphae and normally is club shaped. Two or more basidiospores are produced by the basidium, and basidia may be held within fruiting bodies called **basidiocarps**.

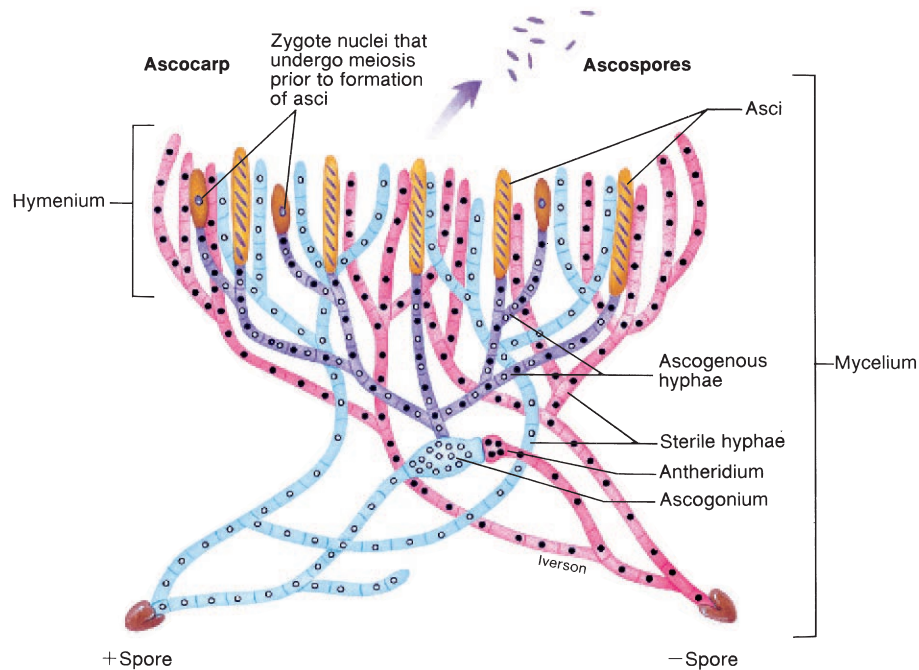
The basidiomycetes affect humans in many ways. Most are saprophytes that decompose plant debris, especially cellulose and lignin. Many mushrooms are used as food throughout the world. The cultivation of *Agaricus campestris* is a multimillion-dollar business (*see figure 41.24*).

Many mushrooms produce specific alkaloids that act as either poisons or hallucinogens. One such example is the "destroying angel" mushroom, *Amanita phalloides*. Two toxins isolated from this species are phalloidin and  $\alpha$ -amanitin. Phalloidin primarily attacks liver cells where it binds to plasma membranes, causing them to rupture and leak their contents. Alpha-amanitin attacks the cells lining the stomach and small intestine and is responsible for the severe gastrointestinal symptoms associated with mushroom poisoning.

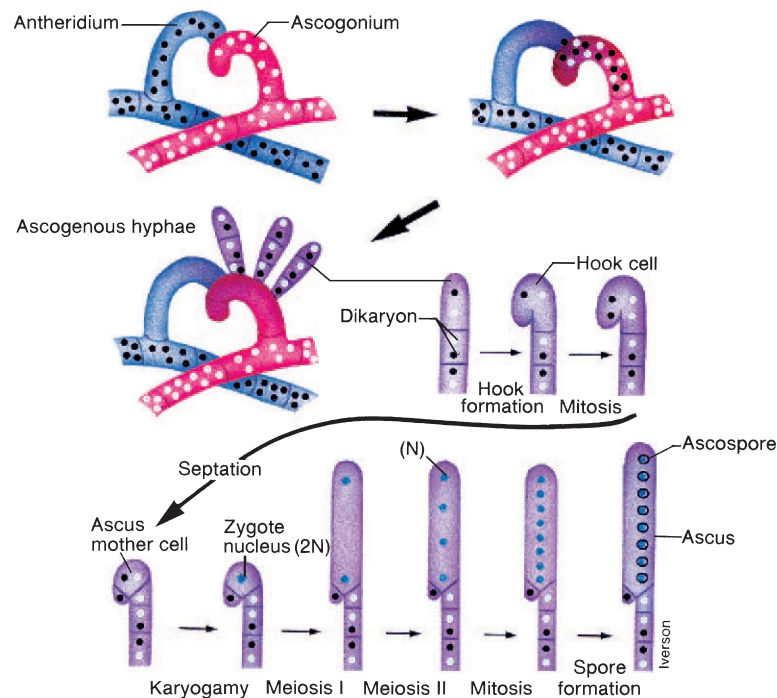
The basidiomycete *Cryptococcus neoformans* is an important human pathogen. It produces the disease called **cryptococcosis**, a systemic infection primarily involving the lungs and central nervous system. Other basidiomycetes, the smuts and rusts, are virulent plant pathogens that cause extensive damage to cereal crops; millions of dollars worth of crops are destroyed annually. In these fungi, large basidiocarps are not formed. Instead the small basidia arise from hyphae at the surface of the host plant. The mycelia grow either intra- or extracellularly in plant tissue.

[Fungal diseases \(pp. 942–50\)](#)



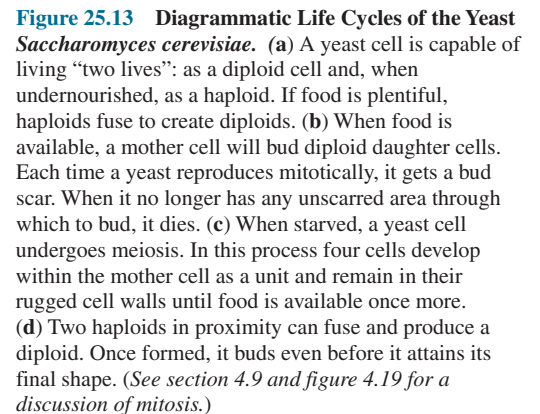


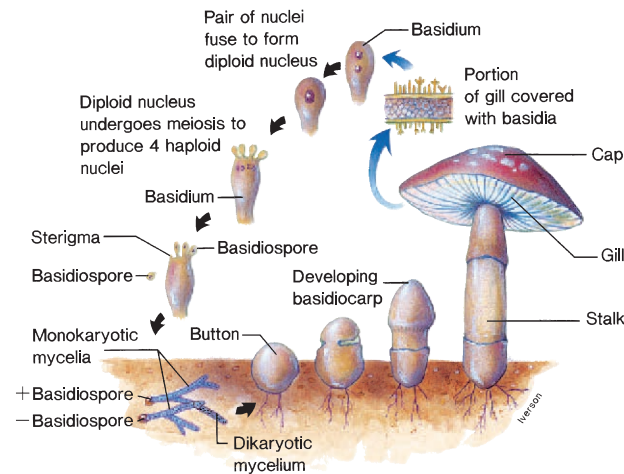
(a)



(b)

**Figure 25.12 The Life Cycle of Ascomycetes.** Sexual reproduction involves the formation of asci and ascospores. Within the ascus, karyogamy is followed by meiosis to produce the ascospores. (a) Sexual reproduction and ascocarp morphology of a cup fungus. (b) The details of sexual reproduction in ascogenous hyphae. The nuclei of the two mating types are represented by unfilled and filled circles. See text for details.





**Figure 25.14 Division Basidiomycota.** The life cycle of a typical soil basidiomycete starts with a basidiospore germinating to produce a monokaryotic mycelium (one with a single nucleus in each septate cell). The mycelium quickly grows and spreads throughout the soil. When this primary mycelium meets another monokaryotic mycelium of a different mating type, the two fuse to initiate a new dikaryotic secondary mycelium. The secondary mycelium is divided by septa into cells, each of which contains two nuclei, one of each mating type. This dikaryotic mycelium is eventually stimulated to produce basidiocarps. A solid mass of hyphae forms a button that pushes through the soil, elongates, and develops a cap. The cap contains many platelike gills, each of which is coated with basidia. The two nuclei in the tip of each basidium fuse to form a diploid zygote nucleus, which immediately undergoes meiosis to form four haploid nuclei. These nuclei push their way into the developing basidiospores, which are then released at maturity.

### Division Deuteromycota

To a large degree, classical fungal taxonomy is based on specific patterns of sexual reproduction. When a fungus lacks the sexual phase (perfect stage), or if this phase has not been observed, it is placed within the division *Deuteromycota*, commonly called the Fungi Imperfecti or **deuteromycetes** (“secondary fungi”). Once a perfect stage is observed, the fungus is transferred to its proper division. Molecular systematics places the *Deuteromycota* among their closest relatives in the *Eumycota* (Phylogenetic Diagram 25).

Most Fungi Imperfecti are terrestrial, with only a few being reported from freshwater and marine habitats. The majority are either saprophytes or parasites of plants. A few are parasitic on other fungi.

Many imperfect fungi directly affect human welfare. Several are human pathogens, causing such diseases as athlete’s foot, ringworm, and histoplasmosis (see section 40.1). The chemical activities of many Fungi Imperfecti are important industrially. For example, some species of *Penicillium* (see figure 4.1c,d) synthesize the well-known antibiotics penicillin and griseofulvin. Other species give characteristic aromas to cheeses such as Gorgonzola, Camembert, and Roquefort.

Different species of *Aspergillus* are used to ferment soy sauce and to manufacture citric, gluconic, and gallic acids. *Aspergillus flavus* and *A. parasiticus* produce secondary metabolites, called aflatoxins, that are highly toxic and carcinogenic to animals and humans (see section 41.2). Another group of fungal toxins, the trichothecenes, are strong inhibitors of protein synthesis in eucaryotic cells.

### Division Chytridiomycota

The simplest of the true fungi belong to the division *Chytridiomycota*. This division contains one class, *Chytridiomycetes*, and its members are known familiarly as the **chytrids**. These are simple terrestrial and aquatic fungi that reproduce asexually by forming motile zoospores with single, posterior, whiplash flagella. The entire organism is microscopic in size and may consist of a single cell, a small multinucleate mass, or a true mycelium. Usually chitin is the major constituent of chytrid cell walls. Chytrids are thought to have been derived from a protozoan ancestor having similar flagellation. They are likely to be ancestral to the remaining groups of true fungi (Phylogenetic Diagram, page 553). Chytrids have a variety of life cycles. When sexual reproduction occurs, it results in a zygote that generally becomes a resting spore or sporangium. Some can grow saprophytically on dead organic matter; others are parasites of algae (see figure 29.7), other true fungi, and terrestrial and aquatic plants. Species such as *Allomyces* are used in the study of morphogenesis.

1. Describe how a typical zygomycete reproduces.
2. Give some beneficial uses for zygomycetes.
3. Describe the ascomycete life cycle. How are the ascomycetes important to humans?
4. How do yeasts reproduce sexually? Asexually?
5. Describe the life cycle of a typical basidiomycete.
6. How do some Fungi Imperfecti affect humans?
7. What are chytrids? Discuss their importance.

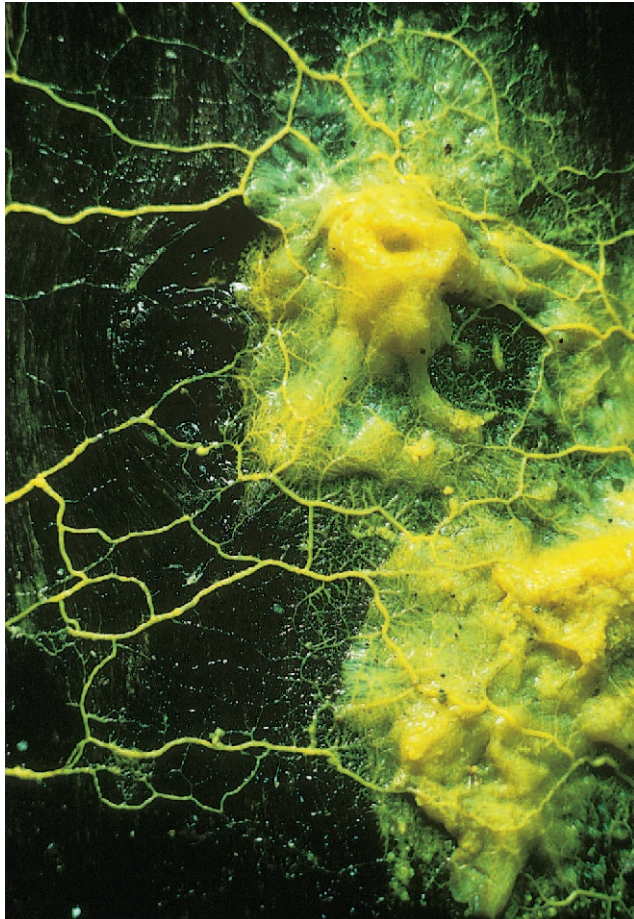
## 25.7 Slime Molds and Water Molds

The **slime molds** and **water molds** resemble fungi in only appearance and life-style. In their cellular organization, reproduction, and life cycles, they are phylogenetically distinct (Phylogenetic Diagram 25).

### Division Myxomycota (Acellular Slime Molds)

Under appropriate conditions **plasmodial (acellular) slime molds** exist as streaming masses of colorful protoplasm that creep along in an amoeboid fashion over moist, rotting logs, leaves, and other organic matter. Feeding is by phagocytosis. Be-





**Figure 25.15** Slime Molds. Plasmodium of the slime mold *Physarum*; light micrograph ( $\times 175$ ).

cause this streaming mass lacks cell walls, it is called a **plasmodium** (figures 25.15 and 25.16a). The plasmodium contains many nuclei, and as the organism grows, the diploid nuclei divide repeatedly.

When the plasmodium matures or when food and/or moisture are scarce, it moves into a lighted area and develops delicate ornate fruiting bodies (figure 25.16b–e). As the fruiting bodies mature, they form spores with cellulose walls that are resistant to environmental extremes. The spores germinate in the presence of adequate moisture to release either nonflagellated amoeboid **myxamoebae** or flagellated **swarm cells**. Initially the myxamoebae or swarm cells feed and are haploid (figure 25.16a); eventually they fuse to form a diploid zygote. The zygote feeds, grows, and multiplies its nuclei through synchronous mitotic division to form the multinucleate plasmodium.

### Division Acrasiomycota (Cellular Slime Molds)

The vegetative stage of **cellular slime molds** consists of individual amoeboid cells termed myxamoebae (figure 25.17a). The myxamoebae feed phagocytically on bacteria and yeasts. When food is plentiful, they divide repeatedly by mitosis and cytokinesis, producing new daughter myxamoebae. As their food supply is exhausted, the myxamoebae begin to secrete cyclic adenosine monophosphate (cAMP). This attracts other myxamoebae that move toward the cAMP chemotactic source and in turn secrete more cAMP. When the individual myxamoebae aggregate (figure 25.17b), they form a sluglike **pseudoplasmodium** (figure 25.17c). The pseudoplasmodium may move around as a unit for a while, leaving a slime trail, but it eventually becomes sedentary. In the culmination of the asexual phase, pseudoplasmodial cells begin to differentiate into prestalk cells and prespore cells (figure 25.17b). A fruiting body called a **sorus** or **sorocarp** forms and matures (figure 25.17d) and then produces spores (figure 25.17e). The spores are eventually released, and when conditions become favorable, they germinate to release haploid amoebae and repeat the cycle.

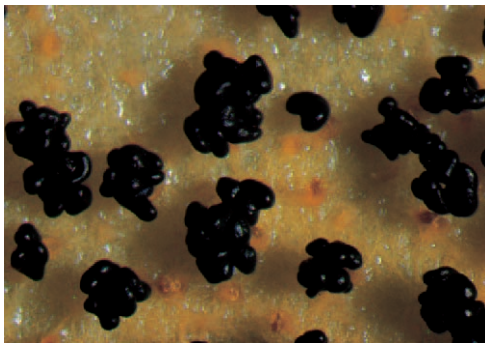
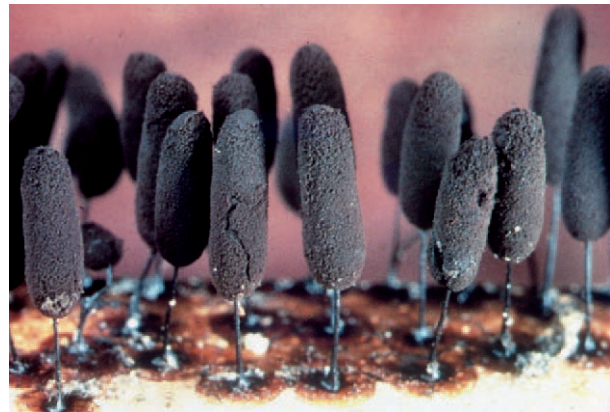
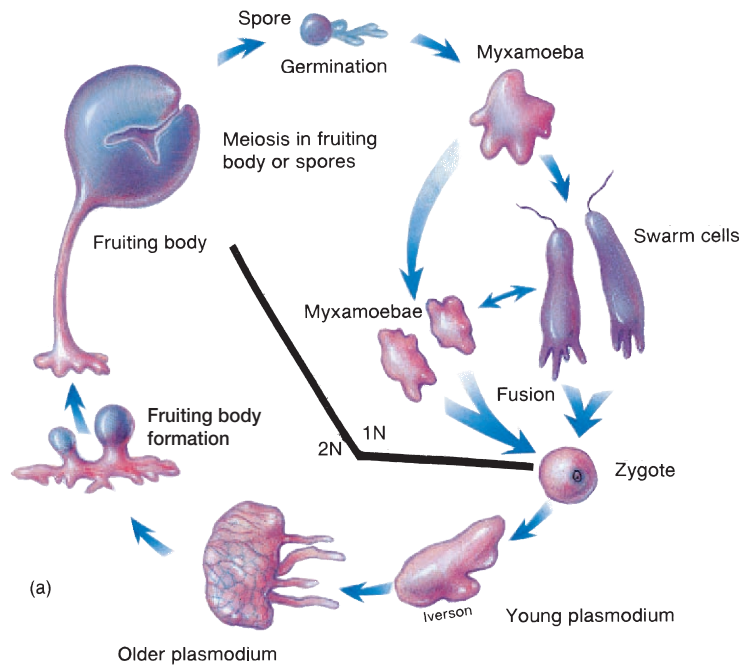
### Division Oomycota

Members of the division *Oomycota* are collectively known as **oomycetes** or water molds. Oomycetes resemble true fungi only in appearance, consisting of finely branched filaments called hyphae. However, oomycetes have cell walls of cellulose, whereas the walls of most fungi are made of chitin. Oomycetes are also unlike the true fungi in that they have tubular mitochondrial cristae.

*Oomycota* means “egg fungi,” a reference to the mode of sexual reproduction in water molds. A relatively large egg cell (oogonium) is fertilized by either a sperm cell or a smaller antheridium to produce a zygote. When the zygote germinates, it forms asexual zoospores that bear flagella.

Water molds such as *Saprolegnia* and *Achlya* are saprophytes that grow as cottony masses on dead algae and small animals, mainly in freshwater environments. They are important decomposers in aquatic ecosystems. Some water molds are parasitic on the gills of fish. The water mold *Peronospora hyoscyami* is currently responsible for the troublesome “blue mold” of tobacco plants throughout the world. In the United States alone, blue mold produces millions of dollars of damage yearly to tobacco crops. Other oomycetes cause late blight of potatoes (*Phytophthora infestans*) and grape downy mildew (*Plasmopara viticola*).

1. What is a plasmodium?
2. How do the plasmodial slime molds reproduce? Describe the cellular slime mold life cycle.
3. Where would you look for an oomycete?

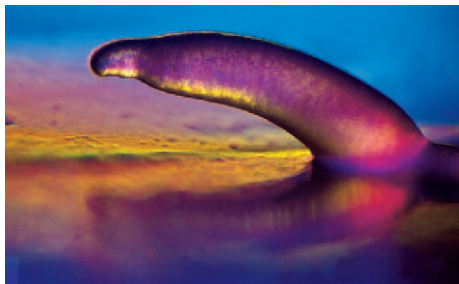


**Figure 25.16** Reproduction in the *Myxomycota*. (a) The life cycle of a plasmodial slime mold. (Different parts of the life cycle are drawn at different magnifications.) Plasmodial slime-mold fruiting bodies: (b) *Hemitrachia* ( $\times 100$ ), (c) *Stemonitis* ( $\times 100$ ), (d) *Physarum polycephalum*, and (e) *Arcyria denudata*.

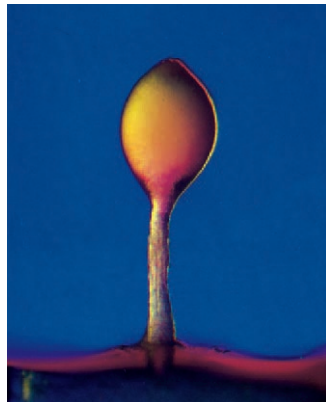




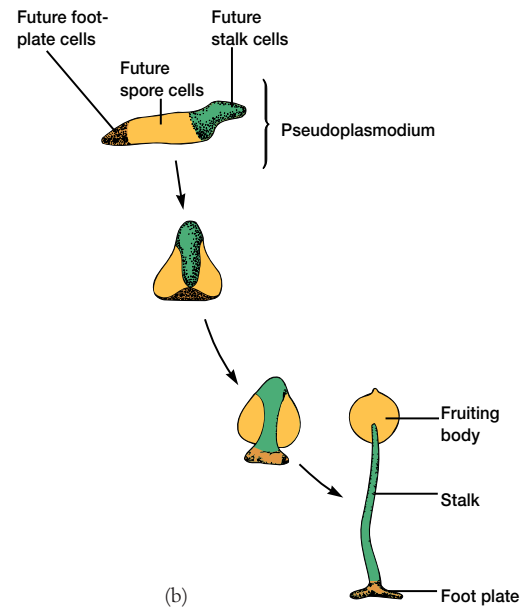
(a)



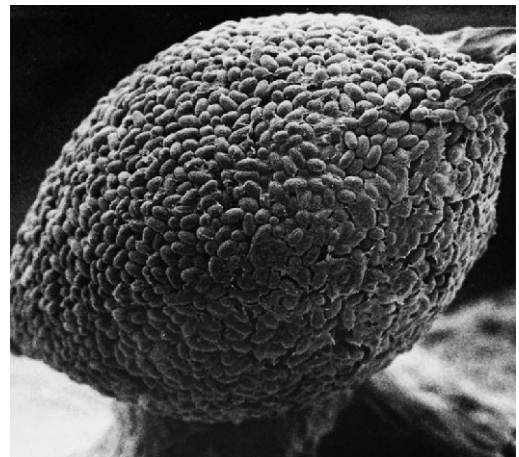
(c)



(d)



(b)



(e)

**Figure 25.17 Division Acrasiomycota.** *Dictyostelium discoideum*, a cellular slime mold. In the free-living stage a myxamoeba resembles an irregularly shaped amoeba. (a) Aggregating myxamoebae become polar and begin to move in an oriented direction due to the influence of cAMP. (b) Diagrammatic drawing of cell migration involved in the formation of the sorocarp from (c) an initial pseudoplasmodium. (d) Light micrograph of a mature fruiting body (sorocarp). (e) Electron micrograph of a sorus showing spores ( $\times 1,800$ ).



## Summary

1. Fungi are omnipresent in the environment, being found wherever water, suitable organic nutrients, and an appropriate temperature occur. They secrete enzymes outside their body structure and absorb the digested food.
2. Fungi are important decomposers that break down organic matter; live as parasites on animals, humans, and plants; play a role in many industrial processes; and are used as research tools in the study of fundamental biological processes.
3. The body or vegetative structure of a fungus is called a thallus (**figure 25.2**). Fungi may be grouped into molds or yeasts based on the development of the thallus.
4. A fungus is a eucaryotic, spore-bearing organism that has absorptive nutrition and lacks chlorophyll; that reproduces asexually, sexually, or by both methods; and that normally has filamentous hyphae surrounded by cell walls, which usually contain chitin.
5. Yeasts are unicellular fungi that have a single nucleus and reproduce either asexually by budding and transverse division or sexually through spore formation (**figure 25.3**).
6. A mold consists of long, branched, threadlike filaments of cells, the hyphae, that form a tangled mass called a mycelium (**figure 25.4**).  
  
Hyphae may be either septate or coenocytic (nonseptate). The mycelium can produce reproductive structures.
7. Some fungi are dimorphic—they alternate between a yeast and a mold form (**table 25.2**).
8. Asexual reproduction often occurs in the fungi by the production of specific types of spores that are easily dispersed (**figure 25.7**).
9. Sexual reproduction is initiated in the fungi by the fusion of hyphae of different mating strains. In some fungi the nuclei in the fused hyphae immediately combine to form a zygote. In others the two genetically distinct nuclei remain separate, forming pairs that divide synchronously. Eventually some nuclei fuse (**figure 25.8**).
10. The zygomycetes are coenocytic. Most are saprophytic. One example is the common bread mold, *Rhizopus stolonifer*. Sexual reproduction occurs through a form of conjugation involving + and – strains (**figure 25.9**).
11. The ascomycetes are known as the sac fungi because they form a sac-shaped reproductive structure called an ascus (**figure 25.10**). In asexual reproduction they produce characteristic conidia (**figure 25.11**). Sexual reproduction involves + and – strains (**figure 25.12**).
12. The basidiomycetes are the club fungi. They are named after their basidium that produces basidiospores (**figure 25.14**).
13. The deuteromycetes (Fungi Imperfecti) are fungi with no known sexual (perfect) phase.
14. The plasmodial (acellular) slime molds move about as a multinucleate plasmodium (**figure 25.15**). When food or moisture is scarce, these slime molds form sporangia within which spores are produced (**figure 25.16**).
15. The cellular slime molds consist of a vegetative stage called a myxamoeba (**figure 25.17**). The myxamoebae feed until their nutrients are exhausted, then the cells come together to form a moldlike multicellular structure called a sorus or sorocarp. The sorocarp produces haploid spores that germinate when conditions are favorable to form new myxamoebae.
16. The chytrids are a group of terrestrial and aquatic fungi that produce motile zoospores with single, posterior, whiplash flagella.
17. The Oomycota (water molds) are characterized by the production of motile spores (zoospores, gametes) and production of resistant sexual spores (oospores).

## Key Terms

|                          |                                       |                      |
|--------------------------|---------------------------------------|----------------------|
| antheridium 561          | conidiospore 557                      | plasmodium 565       |
| arthroconidia 557        | cryptococcosis 561                    | progametangium 560   |
| arthrospore 557          | deuteromycetes 564                    | pseudoplasmodium 565 |
| ascocarp 561             | dikaryotic stage 557                  | saprophyte 557       |
| ascogenous hypha 561     | ergot 561                             | septa 556            |
| ascogonium 561           | ergotism 561                          | septate 556          |
| ascomycetes 560          | Eumycota 553                          | slime mold 564       |
| ascospore 558            | fungus 553                            | sorocarp 565         |
| ascus 561                | gametangium 557                       | sorus 565            |
| basidiocarp 561          | hypha 556                             | sporangiospore 557   |
| basidiomycetes 561       | mold 556                              | sporangium 557       |
| basidiospore 558         | mycelium 556                          | swarm cell 565       |
| basidium 561             | mycologist 553                        | thallus 554          |
| blastospore 557          | mycology 553                          | water mold 564       |
| cellular slime molds 565 | mycosis 553                           | yeast 554            |
| chitin 554               | mycotoxicology 553                    | YM shift 556         |
| chlamydospore 557        | myxamoeba 565                         | zygomycetes 560      |
| chytrids 564             | oomycetes 565                         | zygospore 558        |
| coenocytic 556           | plasmodial (acellular) slime mold 564 |                      |

### Questions for Thought and Review

1. Some fungi can reproduce both sexually and asexually. What are the advantages and disadvantages of each?
2. Why is nutrition in the fungi a property of membrane function?
3. Why are most fungi confined to a specific ecological niche?
4. Some authorities believe that the slime molds and water molds should be placed in the kingdom *Fungi*, whereas others believe that they should be placed in the kingdom *Protista*. What are the characteristics that these organisms share with both the fungi and protists that have led to this ambiguity?
5. At the present time those fungi that have no sexual reproduction cannot be classified with their sexually reproducing relatives. Why? Do you think that this is likely to change in the future?
6. Because asexual spores are such a rapid way of reproducing for some fungi, what adaptive "use" is there for an additional sexual phase?
7. Some fungi can be viewed as coenocytic organisms that exhibit little differentiation. When differentiation does occur, such as in the formation of reproductive structures, it is preceded by septum formation. Why does this occur?
8. The term mushrooming is a proverbial description for expanding rapidly. Why is this an accurate metaphor?
9. Both bacteria and fungi are major environmental decomposers. Obviously competition exists in a given environment, but fungi usually have an advantage. What is this advantage?
10. Very few antibiotics can control fungi, while there are many for bacterial control. Why is this so?

### Critical Thinking Questions

1. What are some logical targets to exploit in treating animals or plants suffering from fungal infections? Are they different from the targets you would use when treating infections caused by bacteria? By viruses? Explain.
2. Fungi tend to reproduce asexually when nutrients are plentiful and conditions are favorable for growth, but reproduce sexually when environmental or nutrient conditions are not favorable. Why is this an evolutionarily important and successful strategy?
3. Compare and contrast the development of fruiting bodies by cellular slime molds and by myxobacteria.

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# CHAPTER 26

## The Algae



This photograph shows the giant kelp (*Macrocystis*) from along the coast of California. Kelps are brown algae; giant kelps can grow to more than 60 m long. This photograph shows the stipes, blades, and air bladders. Giant kelp are anchored to the ocean floor by holdfasts.

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  - Euglenophyta* (Euglenoids) 576
  - Chrysophyta* (Golden-Brown and Yellow-Green Algae; Diatoms) 577
  - Phaeophyta* (Brown Algae) 578
  - Rhodophyta* (Red Algae) 578
  - Pyrrhophyta* (Dinoflagellates) 579

### Concepts

1. Most algae are found in freshwater and marine environments; a few grow in terrestrial habitats.
2. The algae are not a single, closely related taxonomic group but, instead, are a diverse, polyphyletic assemblage of unicellular, colonial, and multicellular eucaryotic organisms.
3. Although algae can be autotrophic or heterotrophic, most are photoautotrophs. They store carbon in a variety of forms, including starch, oils, and various sugars.
4. The body of an alga is called the thallus. Algal thalli range from small solitary cells to large, complex multicellular structures.
5. Algae reproduce asexually and sexually.
6. The following classical divisions of the algae are discussed: *Chlorophyta* (green algae), *Charophyta* (stoneworts/brittleworts), *Euglenophyta* (euglenoids), *Chrysophyta* (golden-brown and yellow-green algae; diatoms), *Phaeophyta* (brown algae), *Rhodophyta* (red algae), and *Pyrrhophyta* (dinoflagellates).



*The term algae means different things to different people, and even the professional botanist and biologist find algae embarrassingly elusive of definition. Thus, laymen have given such names as “pond scums,” “frog spittle,” “water mosses,” and “seaweeds,” while some professionals shrink from defining them.*

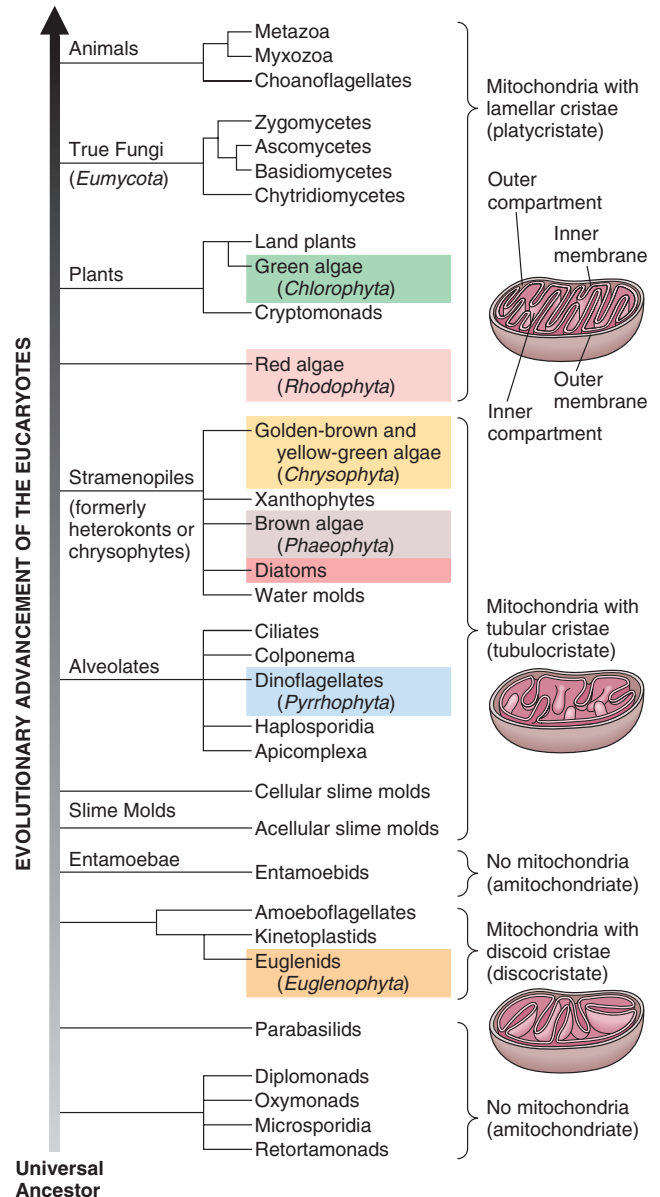
—Harold C. Bold and Michael J. Wynne

Chapter 26 presents some general features of algae. Because 18S rRNA analysis has shown that these organisms arose independently on different occasions, the algae do not represent a monophyletic group (**Phylogenetic Diagram 26**). Accordingly, the taxa “algae” should not be used in molecular taxonomy schemes. The term algae can still be used (as it is in this chapter) to denote a group of related eucaryotic organisms that share some morphological, reproductive, ecological, and biochemical characteristics.

**Phycology** or **algology** is the study of algae. The word phycology is derived from the Greek *phykos*, meaning seaweed. The term **algae** [s., alga] was originally used to define simple “aquatic plants.” As noted above, it no longer has any formal significance in classification schemes. Instead the algae can be described as eucaryotic organisms that have chlorophyll *a* and carry out oxygen-producing photosynthesis. They differ from other photosynthetic eucaryotes in lacking a well-organized vascular conducting system and in having very simple reproductive structures. In sexual reproduction the whole organism may serve as a gamete; unicellular structures (gametangia) may produce gametes; or gametes can be formed by multicellular gametangia in which every cell is fertile. Unlike the case with plants, algal gametangia do not have nonfertile cells.

## 26.1 Distribution of Algae

Algae most commonly occur in water (fresh, marine, or brackish) in which they may be suspended (**planktonic**) or attached and living on the bottom (**benthic**). A few algae live at the water-atmosphere interface and are termed **neustonic**. **Plankton** [Greek *plankos*, wandering] consists of free-floating, mostly microscopic aquatic organisms. **Phytoplankton** is made up of algae and small plants, whereas **zooplankton** consists of animals and nonphotosynthetic protists. Some algae grow on moist rocks, wood, trees, and on the surface of moist soil. Algae also live as endosymbionts in various protozoa, mollusks, worms, and corals. Several algae grow as endosymbionts within plants, some are attached to the surface of various structures, and a few lead a parasitic existence. Algae also associate with fungi to form lichens. *Zooxanthella symbiosis* (p. 599); *Lichen symbiosis* (pp. 598–99)



**Phylogenetic Diagram 26** Tentative Phylogeny of the Algae-like Eucaryotes Based on 18S rRNA Sequence Comparisons. Using molecular systematics, organisms are grouped together based on the molecular phylogeny of their nuclear SSU rRNA genes and the type of mitochondrial cristae present. Accordingly, the algae-like eucaryotic organisms have arisen independently on five different occasions and are polyphyletic (highlighted by different colors).

## 26.2 Classification of Algae

According to the five-kingdom system of Whittaker, the algae belong to seven divisions distributed between two different kingdoms (**table 26.1**). This classical classification is based on

cellular, not organismal, properties. Some more important properties include: (1) cell wall (if present) chemistry and morphology; (2) form in which food or assimilatory products of photosynthesis are stored; (3) chlorophyll molecules and accessory pigments that contribute to photosynthesis; (4) flagella number and the location of their insertion in motile cells; (5) morphology of the cells and/or body (thallus); (6) habitat; (7) reproductive structures; and (8) life history patterns. Based on these properties the algae are arranged by divisions in **table 26.2**, which summarizes their more significant characteristics.

**Table 26.1** Classical Classification of Algae<sup>a</sup>

| Division (Common Name)                                            | Kingdom                                               |
|-------------------------------------------------------------------|-------------------------------------------------------|
| <i>Chrysophyta</i> (yellow-green and golden-brown algae; diatoms) | <i>Protista</i> (single cell or colonial; eucaryotic) |
| <i>Euglenophyta</i> (photosynthetic euglenoid flagellates)        | <i>Protista</i>                                       |
| <i>Pyrrophyta</i> (dinoflagellates)                               | <i>Protista</i>                                       |
| <i>Charophyta</i> (stoneworts)                                    | <i>Protista</i>                                       |
| <i>Chlorophyta</i> (green algae)                                  | <i>Protista</i>                                       |
| <i>Phaeophyta</i> (brown algae)                                   | <i>Plantae</i> (multicellular; eucaryotic)            |
| <i>Rhodophyta</i> (red algae)                                     | <i>Plantae</i>                                        |

<sup>a</sup>Five-kingdom system.

Molecular systems (Phylogenetic Diagram 26) have placed some of the classical algae with plants (green algae); some as a separate lineage (red algae); some with the stramenopiles (golden-brown and yellow-green algae, brown algae, and diatoms); some with the alveolates (dinoflagellates); and still others with some protozoa (euglenoids). Two of these groups, the alveolates and stramenopiles, have been created recently as a result of rRNA comparisons and ultrastructural studies. The alveolates have mitochondria with tubular cristae and subsurface alveoli or sacs that abut the surface. Dinoflagellates, ciliate protozoa, and the apicomplexan protozoa are alveolates (Phylogenetic Diagram 26). The stramenopiles have mitochondria with tubular cristae and hollow hairs that give rise to a small number of fine hairs (tripartite tubular hairs). These hairs are usually on their flagella. Photosynthetic forms often have chlorophylls a and c. Some common stramenopiles are the opalinid protozoa, oomycetes, diatoms, brown algae or phaeophytes, chrysophytes, and xanthophytes. Although a few groups such as the diatoms have lost their hairs, they are still considered stramenopiles based on the rRNA data, mitochondrial characteristics, and other properties.

### 26.3 Ultrastructure of the Algal Cell

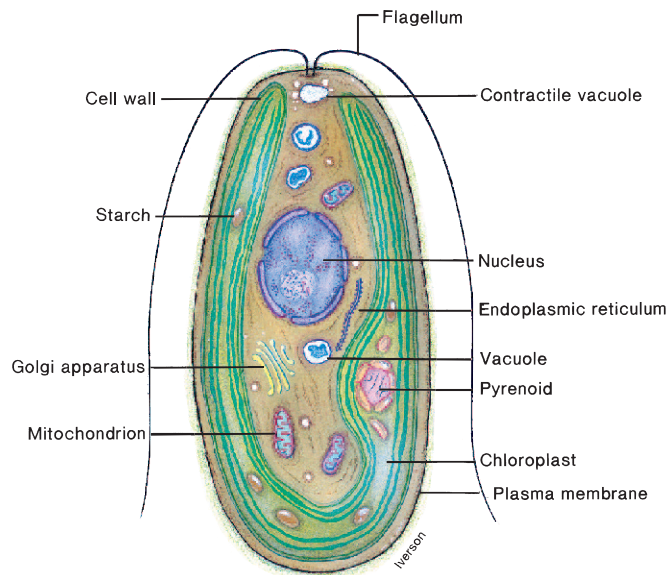
The eucaryotic algal cell (**figure 26.1**) is surrounded by a thin, rigid cell wall. Some algae have an outer matrix lying outside the cell wall. This usually is flexible and gelatinous, similar to bacterial capsules. When present, the flagella are the locomotor organelles. The nucleus

**Table 26.2** Comparative Summary of Some Algal Characteristics

| Division            | Approximate Number of Species | Common Name and Representative                                      | Pigments                                                           |                                                     |                                                               | Thylakoids per Stack in Chloroplast |
|---------------------|-------------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------|---------------------------------------------------------------|-------------------------------------|
|                     |                               |                                                                     | Chlorophylls                                                       | Phycobilins (Phycobiliproteins)                     | Carotenoids                                                   |                                     |
| <i>Chlorophyta</i>  | 7,500                         | Green algae<br>( <i>Chlamydomonas</i> )                             | <i>a, b</i>                                                        | —                                                   | β-carotene,<br>± α-carotene,<br>xanthophylls                  | 3–6                                 |
| <i>Charophyta</i>   | 250                           | Stoneworts or brittleworts<br>( <i>Chara</i> )                      | <i>a, b</i>                                                        | —                                                   | α-, β-, τ-carotene,<br>xanthophylls                           | Many                                |
| <i>Euglenophyta</i> | 700                           | Euglenoids<br>( <i>Euglena</i> )                                    | <i>a, b</i>                                                        | —                                                   | β-carotene,<br>xanthophylls,<br>± τ-carotene                  | 3                                   |
| <i>Chrysophyta</i>  | 6,000                         | Golden-brown, yellow-green algae; diatoms,<br>( <i>Cyclotella</i> ) | <i>a, c</i> <sub>1</sub> / <i>c</i> <sub>2</sub> , rarely <i>d</i> | —                                                   | α-, β-, ε-carotene,<br>fucoxanthin,<br>xanthophylls           | 3                                   |
| <i>Phaeophyta</i>   | 1,500                         | Brown algae<br>( <i>Sargassum</i> )                                 | <i>a, c</i>                                                        | —                                                   | β-carotene, fucoxanthin,<br>xanthophylls                      | 3                                   |
| <i>Rhodophyta</i>   | 3,900                         | Red algae<br>( <i>Corallina</i> )                                   | <i>a</i> , rarely <i>d</i>                                         | C-phycocyanin,<br>allophycocyanin,<br>phycoerythrin | Xanthophylls<br>(β-carotene,<br>zeaxanthine,<br>± α-carotene) | 1                                   |
| <i>Pyrrophyta</i>   | 1,100                         | Dinoflagellates<br>( <i>Gymnodinium</i> )                           | <i>a, c</i> <sub>1</sub> , <i>c</i> <sub>2</sub>                   | —                                                   | β-carotene, fucoxanthin,<br>peridinin, dinoxanthin            | 3                                   |

<sup>a</sup> Refers specifically to the vegetative cells. Spores, akinetes, zygotes contain waxes, nonsaponifiable polymers, and phenolic substances.

<sup>b</sup> The following abbreviations are used: fresh water (fw), brackish water (bw), salt water (sw), terrestrial (t).



**Figure 26.1 Algal Morphology.** Schematic drawing of a typical eucaryotic algal cell showing some of its organelles and other structures.

has a typical nuclear envelope with pores; within the nucleus are a nucleolus, chromatin, and karyolymph. The chloroplasts have membrane-bound sacs called thylakoids that carry out the light reactions of photosynthesis. These organelles are embedded in the stroma where the dark reactions of carbon dioxide fixation take place. A dense proteinaceous area, the **pyrenoid** that is associated with synthesis and storage of starch may be present in the chloroplasts.

Mitochondrial structure varies greatly in the algae. Some algae (euglenoids) have discoid cristae; some, lamellar cristae (green and red algae); and the remaining, (golden-brown and yellow-green, brown, and diatoms) have tubular cristae.

1. Define the word algae.
2. To what two kingdoms do the algae belong? How do the classical and molecular classification systems differ?
3. What are some general characteristics of an algal cell?

## 26.4 Algal Nutrition

Algae can be either autotrophic or heterotrophic. Most are photoautotrophic; they require only light and  $\text{CO}_2$  as their principal source of energy and carbon. Chemoheterotrophic algae require external organic compounds as carbon and energy sources. [Types of microbial nutrition](#) (pp. 97–98); [Photosynthesis](#) (pp. 195–99, 207–8)

## 26.5 Structure of the Algal Thallus (Vegetative Form)

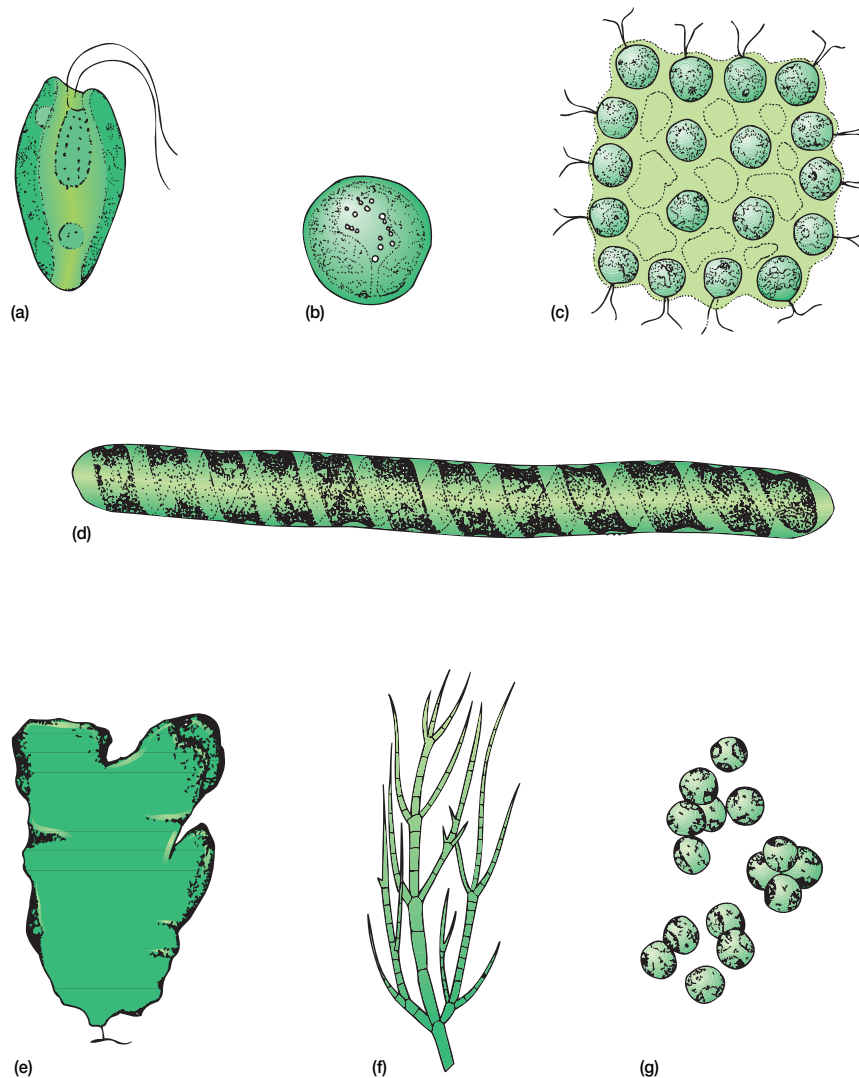
The vegetative body of algae is called the **thallus** [pl., thalli]. It varies from the relative simplicity of a single cell to the more striking complexity of multicellular forms, such as the giant kelps. Single-celled algae may be as small as bacteria, whereas kelp can attain a size over 75 m in length. Algae are unicellular (**figure 26.2a,b,g**), colonial (**figure 26.2c**), filamentous (**figure 26.2d**), membranous and bladelike (**figure 26.2e**), or tubular (**figure 26.2f**).

## 26.6 Algal Reproduction

Some unicellular algae reproduce asexually. In this kind of reproduction, gametes do not fuse to form a zygote. There are three basic types of asexual reproduction: fragmentation, spores, and binary fission. In **fragmentation** the thallus breaks up and each fragmented part grows to form a new thallus. **Spores** can be formed in ordinary vegetative cells or in specialized structures termed sporangia [s., sporangium; Greek *spora*, seed, and *angeion*, vessel]. Flagellated motile spores are called **zoospores**. Nonmotile spores produced by sporangia are termed **aplanospores**. In some unicellular algae **binary fission** occurs (nuclear division followed by division of the cytoplasm).

| Storage Products                          | Flagella                               | Cell Wall <sup>a</sup>                                 | Habitat <sup>b</sup> |
|-------------------------------------------|----------------------------------------|--------------------------------------------------------|----------------------|
| Sugars, starch, fructosan                 | 1, 2–8; equal, apical or subapical     | Cellulose, mannan, protein, $\text{CaCO}_3$            | fw, bw, sw, t        |
| Starch                                    | 2; subapical                           | Cellulose, $\text{CaCO}_3$                             | fw, bw               |
| Paramylon, oils, sugars                   | 1–3; slightly apical                   | Absent                                                 | fw, bw, sw, t        |
| Chrysolaminarin, oils                     | 1–2; equal or unequal, apical; or none | Cellulose, silica, $\text{CaCO}_3$ , chitin, or absent | fw, bw, sw, t        |
| Laminarin, mannitol, oils                 | 2; unequal, lateral                    | Cellulose, alginic acid, fucoidan                      | bw, sw               |
| Glycogenlike starch (floridean glycoside) | Absent                                 | Cellulose, xylans, galactans, $\text{CaCO}_3$          | fw, bw, sw           |
| Starch, glucan, oils                      | 2; 1 trailing, 1 girdling              | Cellulose, or absent                                   | fw, bw, sw           |





**Figure 26.2 Diagrammatic Algal Bodies:** (a) unicellular, motile, *Cryptomonas*; (b) unicellular, nonmotile, *Palmellopsis*; (c) colonial, *Gonium*; (d) filamentous, *Spirotaenia*; (e) bladelike kelp, *Monostroma*; (f) leafy tubular axis, branched tufts or plumes, *Stigeoclonium*; (g) unicellular, nonmotile, *Chrysocapsa*.

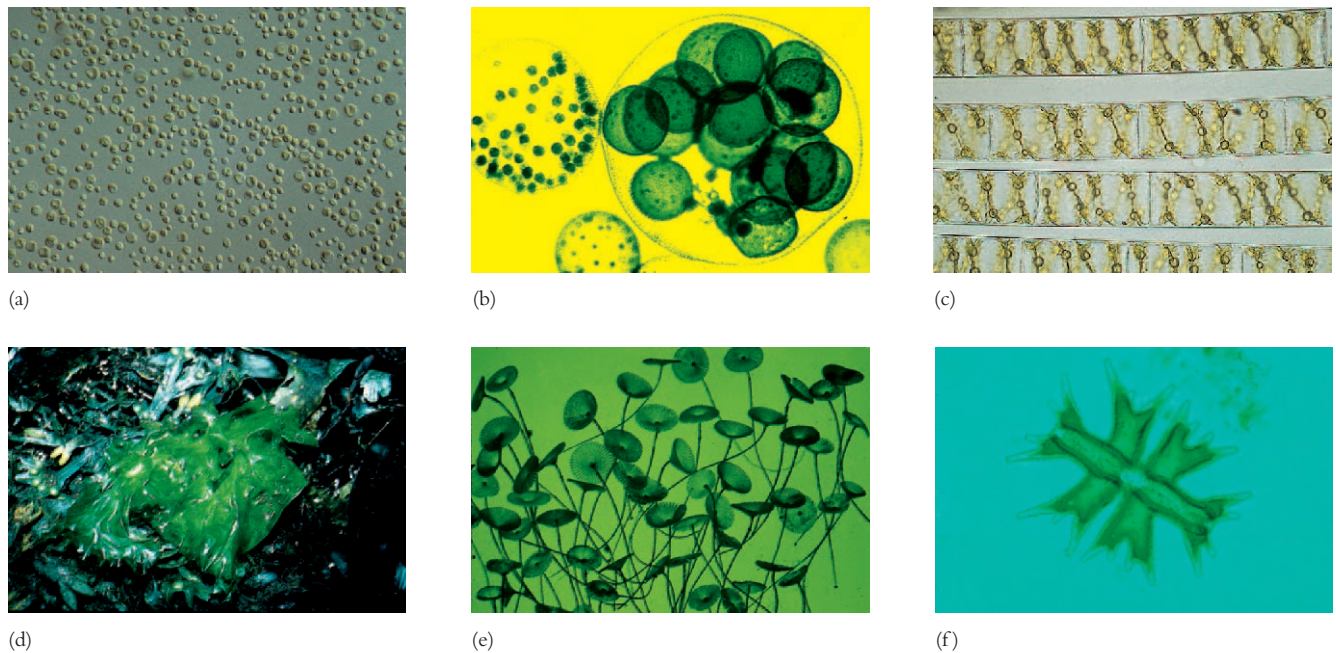
Other algae reproduce sexually. Eggs are formed within relatively unmodified vegetative cells called **oogonia** [s., oogonium] that function as female structures. Sperm are produced in special male reproductive structures called **antheridia** [s., antheridium]. In sexual reproduction these gametes fuse to produce a diploid **zygote**.

1. How can algae be classified based on their mode of nutrition?
2. What are the different types of algal thalli?
3. How do algae reproduce asexually?
4. How do algae reproduce sexually?

## 26.7 Characteristics of the Algal Divisions

### *Chlorophyta* (Green Algae)

The *Chlorophyta* or green algae [Greek *chloros*, green] are an extremely varied division. They grow in fresh and salt water, in soil, on other organisms, and within other organisms. The *Chlorophyta* have chlorophylls a and b along with specific carotenoids, and they store carbohydrates as starch. Many have cell walls of cellulose. They exhibit a wide diversity of body forms, ranging from unicellular to colonial, filamentous, membranous or sheetlike, and tubular types (figure 26.3). Some species have a holdfast structure



**Figure 26.3** *Chlorophyta* (Green Algae); Light Micrographs. (a) *Chlorella*, a unicellular nonmotile green alga ( $\times 160$ ). (b) *Volvox*, a typical green algal colony ( $\times 450$ ). (c) *Spirogyra*, a filamentous green alga ( $\times 100$ ). Four filaments are shown. Note the ribbonlike, spiral chloroplasts within each filament. (d) *Ulva*, commonly called sea lettuce, has a leafy appearance. (e) *Acetabularia*, the mermaid's wine goblet. (f) *Micrasterias*, a large desmid ( $\times 150$ ).

that anchors them to the substratum. Both asexual and sexual reproduction occur in green algae. In molecular classification schemes, green algae are associated with the land plants and have mitochondria with lamellar cristae (**Phylogenetic Diagram 26**).

*Chlamydomonas* is a representative unicellular green alga (figure 26.4). Individuals have two flagella of equal length at the anterior end by which they move rapidly in water. Each cell has a single haploid nucleus, a large chloroplast, a conspicuous pyrenoid, and a **stigma (eyespot)** that aids the cell in phototactic responses. Two small contractile vacuoles at the base of the flagella function as osmoregulatory organelles that continuously remove water. *Chlamydomonas* reproduces asexually by producing zoospores through cell division. The alga also reproduces sexually when some products of cell division act as gametes and fuse to form a four-flagellated diploid zygote that ultimately loses its flagella and enters a resting phase. Meiosis occurs at the end of this resting phase and produces four haploid cells that give rise to adults.

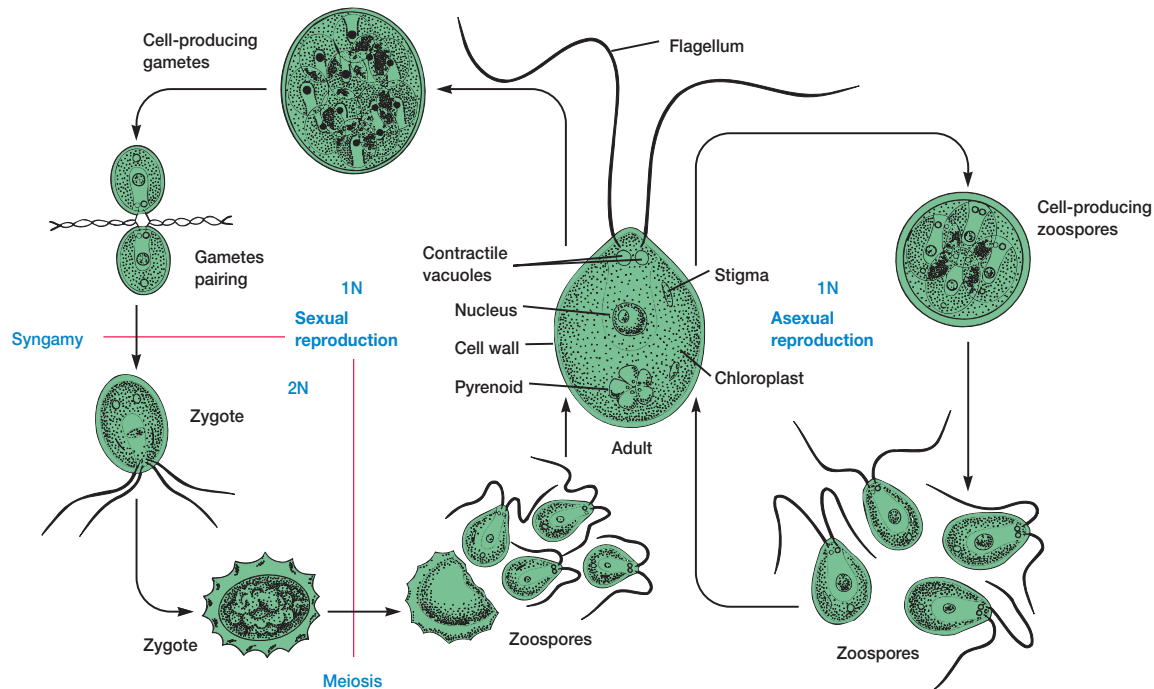
From organisms like *Chlamydomonas*, several distinct lines of evolutionary specialization have evolved in the green algae. The first line contains nonmotile unicellular green algae, such as *Chlorella*. *Chlorella* (figure 26.3a) is widespread both in fresh and salt water and also in soil. It only reproduces asexually and lacks flagella, eyespots, and contractile vacuoles; the nucleus is very small.

Motile, colonial organisms such as *Volvox* represent a second major line of evolutionary specialization. A *Volvox* colony (figure

26.3b; see also figure 2.8b) is a hollow sphere made up of a single layer of 500 to 60,000 individual cells, each containing two flagella and resembling a *Chlamydomonas* cell. The flagella of all the cells beat in a coordinated way to rotate the colony in a clockwise direction as it moves through the water. Only a few cells are reproductive, and these are located at the posterior end of the colony. Some divide asexually and produce new colonies. Others produce gametes. After fertilization, the zygote divides to form a daughter colony. In both cases the daughter colonies stay within the parental colony until it ruptures.

A green alga, *Prototheca moriformis*, causes the disease **protothecosis** in humans and animals. *Prototheca* cells are fairly common in the soil, and it is from this site that most infections occur. Severe systemic infections, such as massive invasion of the bloodstream, have been reported in animals. More common in humans is the subcutaneous type of infection. It starts as a small lesion and spreads slowly through the lymph glands, covering large areas of the body.

1. What body forms do the green algae exhibit?
2. How do green algae reproduce?
3. Describe the structure of *Chlamydomonas*, *Chlorella*, and *Volvox*.
4. Why is the green alga, *Prototheca*, medically important?



**Figure 26.4** *Chlamydomonas*: The Structure and Life Cycle of This Motile Green Alga. During asexual reproduction, all structures are haploid; during sexual reproduction, only the zygote is diploid.

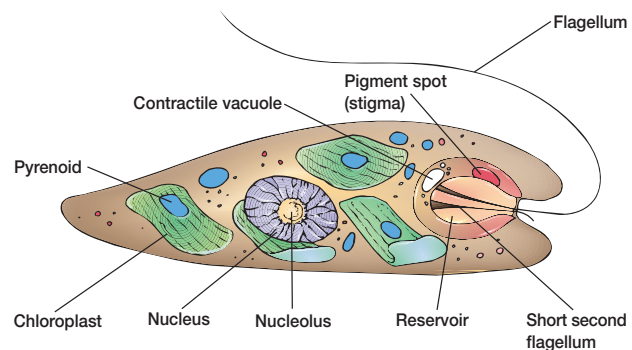
### Charophyta (Stoneworts/Brittleworts)

The **stoneworts** are abundant in fresh to brackish waters and have a worldwide distribution. Often they appear as a dense covering on the bottom of shallow ponds. Some species precipitate calcium and magnesium carbonate from the water to form a limestone covering, thus giving the *Charophyta* their common names of stoneworts or brittleworts.

### Euglenophyta (Euglenoids)

The **euglenoids** share with the *Chlorophyta* and *Charophyta* the presence of chlorophylls *a* and *b* in their chloroplasts. The primary storage product is paramylon (a polysaccharide composed of  $\beta$ -1,3 linked glucose molecules), which is unique to euglenoids. They occur in fresh, brackish, and marine waters and on moist soils; they often form water blooms in ponds and cattle water tanks. In molecular classification schemes, euglenoids are associated with the amoeboflagellates (flagellated protozoa) and kinetoplastids because all members have related rRNA sequences and mitochondria with discoid cristae at some stage in their life cycle (Phylogenetic Diagram 26).

The representative genus is *Euglena*. A typical *Euglena* cell (figure 26.5) is elongated and bounded by a plasma membrane. Inside the plasma membrane is a structure called the **pellicle**, which is composed of articulated proteinaceous strips lying side by side. The pellicle is elastic enough to enable turning and flex-



**Figure 26.5** *Euglena*. A Diagram Illustrating the Principal Structures Found in This Euglenoid. Notice that a short second flagellum does not emerge from the anterior invagination. In some euglenoids both flagella are emergent.

ing of the cell, yet rigid enough to prevent excessive alterations in shape. The several chloroplasts contain chlorophylls *a* and *b* together with carotenoids. The large nucleus contains a prominent nucleolus. The stigma is located near an anterior reservoir. A large contractile vacuole near the reservoir continuously collects water from the cell and empties it into the reservoir, thus regulating the osmotic pressure within the organism. Two flagella arise from the base of the reservoir, although only one emerges from